DATASET-DRIVEN CHANNEL MASKS IN TRANSFORMERS FOR MULTIVARIATE TIME SERIES

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ABSTRACT

Capturing channel dependency (CD) is essential for modeling multivariate time series (TS), and attention-based methods have been widely employed for this purpose. Nonetheless, these methods primarily focus on modifying the *architecture*, often neglecting the importance of *dataset-specific* characteristics. In this work, we introduce the concept of **partial channel dependence** (PCD) to enhance CD modeling in Transformer-based models by leveraging *dataset-specific information* to refine the CD captured by the model. To achieve PCD, we propose **channel masks** (CMs), which are integrated into the attention matrices of Transformers via element-wise multiplication. CMs consist of two components: 1) a **similarity matrix** that captures relationships between the channels, and 2) dataset-specific and learnable **domain parameters** that refine the similarity matrix. We validate the effectiveness of PCD across diverse tasks and datasets with various backbones. Code is available at this repository: `https://github.com/YonseiML/pcd`.

***Index Terms*—** Time Series, Transformer, Channel Dependence, Time Series Forecasting & Classification

1 Introduction

Multivariate time series (MTS) consist of multiple interrelated channels and are prevalent in various real-world applications [1]. MTS forecasting has been explored with two different strategies: the *channel-dependent* (CD) strategy [2, 3, 4] and the *channel-independent* (CI) strategy [5, 6, 7, 8], with the former emphasizing inter-channel dependencies, while the latter ignoring these dependencies and dealing with channels individually. These two strategies are known to have a capacity-robustness trade-off [9], where CD models offer higher capacity but lower robustness, and CI models do vice versa.

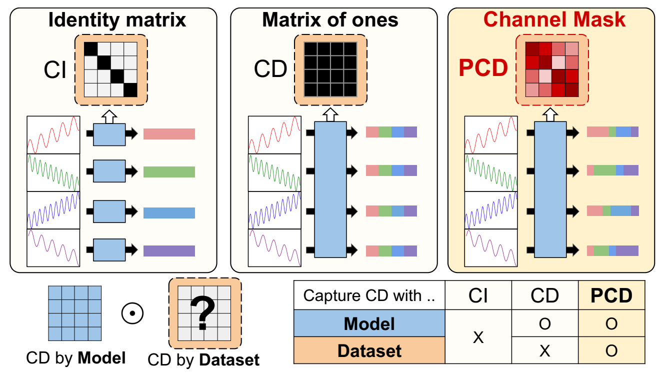

Fig. 1: **CI vs. CD vs. PCD framework.** Under the **partial channel dependence (PCD)** framework, CD captured by model is adjusted with **channel mask** (i.e., CD captured by dataset).

Recently, with the rise of iTransformer [10], various methods [11, 2] have adopted attention mechanisms [12] to capture CD. Furthermore, with the emergence of large-scale TS datasets [13], the CD strategy proves more effective than the CI strategy due to the capacity-robustness trade-off [9], where the higher capacity of CD models benefits larger datasets. In line with these trends, *we highlight the importance of CD*, as CI can be derived from CD by ignoring unnecessary dependencies when well captured. Nonetheless, previous CD methods [11, 14, 15, 16, 17] have focused primarily on the *model architecture*, neglecting the importance of the *dataset*.

To this end, we introduce the concept of **partial channel dependence** (PCD), which improves CD modeling in Transformer-based models by incorporating *dataset-specific* information. Specifically, we propose a **channel mask** (CM) to achieve PCD, a *dataset-specific* matrix that adjusts the CD captured by the model through element-wise multiplication with the attention matrix, as shown in Figure 1 and Figure 2 (a). A CM containing dataset-specific information consists of 1) a **similarity matrix** between the channels, calculated from the *entire dataset* in the data space and 2) **domain parameters** specific to each dataset, which refine the similarity matrix to capture absolute dependencies.

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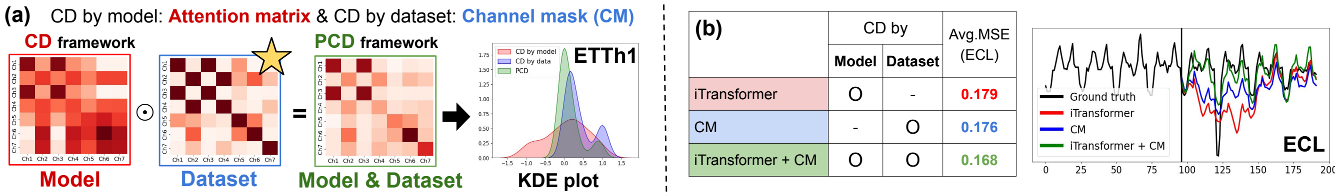

Fig. 2: **Necessity of CD by dataset.** (a) presents CDs captured by model, dataset, and both, along with their distributions, where *CD by model* is adjusted with *CD by dataset* within the PCD framework. (b) shows the TS forecasting results using model (iTransformer), dataset (replacing attention matrix of iTransformer with CM), and both (iTransformer with CM), highlighting the importance of leveraging the *dataset* itself.

Furthermore, we argue that PCD is particularly crucial for TS foundation models (TSFMs) for two reasons: 1) A TSFM is a *single* model trained on *multiple* datasets with varying CD, where each dataset may benefit from a different (CI or CD) approach [13], and 2) a CD architecture (e.g., attention mechanism) is essential for a TSFM due to the capacity-robustness trade-off [9], as it is trained with *large-scale* datasets.

The main contributions are summarized as:

- We introduce the concept of **partial channel dependence** (PCD), where the CD captured by the Transformer-based model is adjusted with the characteristics of the dataset.
- We propose a **channel mask** (CM) to achieve PCD, consisting of 1) a *similarity matrix* between the channels of the entire dataset and 2) *domain parameters* refining the similarity matrix. As a plug-in method, CM is element-wise multiplied with the attention matrix (i.e., CD captured by the model), making it applicable to any model that captures CD with an attention mechanism.
- We present extensive experiments across five backbones, including a TSFM pretrained on multiple datasets, showing consistent performance gains. For instance, applying CMs to iTransformer [10] results in gains across all 13 datasets, yielding an average MSE improvement of 6.3%.

2 Related Works

MTS forecasting models. For CI models, DLinear [5] and RLinear [6] employ linear models along the time dimension, PatchTST [7] divides TS into patches and feeds them into a Transformer in a CI manner, and PITS [8] combines CI and patch independent architectures with multi-layer perceptrons (MLPs). For CD models, Crossformer [15] captures both temporal dependencies (TD) and CD using an attention mechanism, TSMixer [16] utilizes MLPs with patching to capture both dependencies, and CrossGNN [17] employs a linear complexity graph neural network to capture CD. Beyond these methods, various methods have underscored the importance of capturing CD, where LIFT [3] captures the lead-lag relationship between channels and CDAM [4] minimizes redundant information while enhancing relevant mutual information between channels.

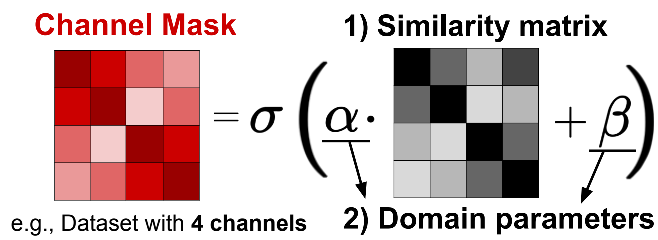

Fig. 3: **Channel Mask.** CM consists of 1) a *similarity matrix* between channels and 2) *domain parameters* to refine the similarity matrix.

Transformer-based CD models. Recently, iTransformer [10] inverts the traditional Transformer framework in TS domain by treating each channel as a token instead of each patch. Following this framework, various methods have been proposed to capture CD with attention mechanism. CARD [11] employs channel-aligned attention to capture both TD and CD and PRformer [18] employs RNN and attention to capture TD and CD, respectively. Minusformer [19] applies a Boosting ensemble learning to channel-wise attention. VCformer [14] introduces a variable correlation attention module that modifies the standard attention. While these works capture CDs, they either *rely on input-window correlations* [14] or *require architectural modifications* [17]. Our approach differs by modeling CD at both local (input-dependent) and global (entire dataset level) scales within a general plug-in framework.

3 Methodology

In this section, we introduce a channel mask (CM), a simple yet effective method for achieving PCD. A CM employs a 1) *similarity matrix* to capture CD with the entire dataset in the data space and 2) *domain parameters* that refine the matrix to learn absolute dependencies specific to each dataset.

3.1 Channel Mask: CD by Dataset

As shown in Figure 3, a CM consists of two components: 1) similarity matrix ($\mathbf{R}$) between channels, and 2) domain parameters (α and β), which scale and shift the matrix according to the dataset's characteristics, along with a sigmoid function to normalize the values between 0 and 1.

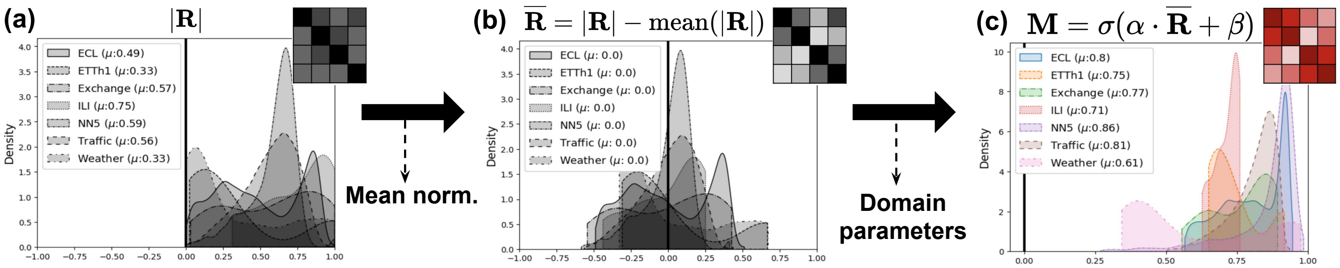

Fig. 4: **Necessity of domain parameters.** As similarity metric is a relative measure depending on the dataset, we employ domain parameters to adjust similarity matrix. Specifically, we refine the matrix with 1) mean normalization and 2) domain parameters, resulting in $\mathbf{M} = \sigma(\alpha \cdot \bar{\mathbf{R}} + \beta)$.

1) Similarity matrix. Correlation measures the relationships between channels and has been used in previous works to analyze CD [14, 3]. Building on these approaches, we employ a correlation matrix ($\mathbf{R}$) as a similarity matrix. However, strong relationhip implies either highly positive or highly negative correlation, as the values of correlation range from -1 to 1, implying no relationship with correlation of 0. To address this issue, we use the absolute value of the matrix $|\mathbf{R}|$. Robustness to the choice of similarity metric is discussed in Table 10.

2) Domain parameters. Although a similarity matrix captures dataset-specific information as prior knowledge using the statistics of a dataset, we argue that the matrix alone might be insufficient for modeling a CM for the following two reasons:

- **a) Relative measure.** Similarity is a dataset-dependent measure, where different datasets exhibit different distributions of elements of $|\mathbf{R}|$, as shown in shown in Figure 4(a).
- **b) Varying dataset characteristics.** The relationship between correlation and CD may *vary across datasets* (i.e., a same correlation can correspond to different levels of CD).

To handle the former (i.e., relative measure), we normalize $|\mathbf{R}|$ by subtracting the mean value, resulting in $\bar{\mathbf{R}}$, as shown in Figure 4(b). To handle the latter (i.e., varying datasets), we introduce two learnable domain parameters (α and β) for each dataset to refine $|\mathbf{R}|$ with affine transformation. Using these parameters along with a sigmoid function, we model a CM as $\mathbf{M} = \sigma(\alpha \cdot \bar{\mathbf{R}} + \beta)$, as shown in Figure 4(c). We argue that using parameters to address the discrepancies is particularly crucial for a TSFM since it is trained on *multiple* datasets, as discussed in Tables 8 and 7.

3.2 CD by Model & CD by Dataset

The proposed CM adjusts the CD captured by the model by performing element-wise multiplication with the attention matrix, with the general adjustment modeled by $\mathbf{A}$:

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\mathbf{A} \odot \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \cdot \mathbf{V},$$

$$\text{where } \mathbf{A} = \begin{cases} \mathbf{I}_{C \times C} & \text{if CI,} \\ \mathbf{1}_{C \times C} & \text{if CD,} \\ \mathbf{M} = \sigma(\alpha \cdot \bar{\mathbf{R}} + \beta) & \text{if PCD,} \end{cases} \tag{1}$$

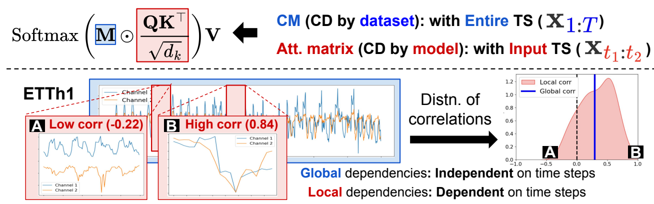

Fig. 5: **Global & local dependencies.** CM and attention matrix are complementary in capturing global and local dependencies, respectively, since CM is constructed from the *entire TS*, while the attention matrix is computed from the *input TS* (segment of the entire TS).

and C is the number of channels. Note that Equation 1 incorporates both CI and CD frameworks within a single expression. As shown in Figure 1, $\mathbf{A}$ is the identity matrix ($\mathbf{I}_{C \times C}$) in the CI framework, while $\mathbf{A}$ is a matrix of ones ($\mathbf{1}_{C \times C}$) in the CD framework. In contrast, PCD framework represents it as $\mathbf{M} = \sigma(\alpha \cdot \bar{\mathbf{R}} + \beta)$, enabling a more refined adjustment of CD tailored to the dataset.

Global & local dependencies. As a similarity matrix is calculated based on the *entire TS*, CM (i.e., CD by dataset), denoted by $\mathbf{M}$, captures the *global* dependencies shared across all time steps. This complements the attention matrix ($\mathbf{Q}\mathbf{K}^\top/\sqrt{d_k}$) (i.e., CD by model), which is calculated based on the *input TS* and thus captures the *local* dependencies which vary by input time step. As shown in Figure 5, PCD framework captures both dependencies through the element-wise multiplication of a CM and an attention matrix. Furthermore, the figure illustrates two channels from ETTh1 [20], showing that even with the existence of global dependencies (red line), the dependency vary across time steps, emphasizing the need to capture both dependencies. Further analysis on the importance of capturing both dependencies is discussed in Table 6.

3.3 Channel Dependence Ratio

To quantify the degree of CD for each dataset trained with the same model, we propose measuring the *channel dependence ratio* (CD ratio), a metric based on a CM. The CD ratio of $\mathbf{M}$, denoted as $r(\mathbf{M})$, is the average of the off-diagonal elements of $\mathbf{M}$, excluding the autocorrelations of their respective channels. This metric yields a value of 0 for CI cases and 1 for CD cases,

Average 4Hs	iTransformer MSE	iTransformer MAE	+ CM MSE	+ CM MAE	Imp. (MSE)	CARD$^{L=96}$ MSE	CARD$^{L=96}$ MAE	+ CM MSE	+ CM MAE	Imp. (MSE)	CARD$^{L=720}$ MSE	CARD$^{L=720}$ MAE	+ CM MSE	+ CM MAE	Imp. (MSE)
ETTh1	0.457	0.449	**0.444**	**0.441**	**2.8%**	0.444	0.429	**0.438**	**0.425**	**1.4%**	0.406	0.428	**0.405**	**0.427**	**0.2%**
ETTh2	0.384	0.407	**0.383**	**0.406**	**0.3%**	0.360	0.387	**0.360**	**0.387**	**0.0%**	0.330	0.378	**0.328**	**0.377**	**0.6%**
ETTm1	0.408	0.412	**0.398**	**0.406**	**2.5%**	0.383	0.382	**0.379**	**0.381**	**1.1%**	0.350	0.370	**0.347**	**0.369**	**0.9%**
ETTm2	0.293	0.337	**0.289**	**0.335**	**1.4%**	0.272	0.314	**0.270**	**0.313**	**0.7%**	0.252	0.309	**0.250**	**0.308**	**0.8%**
PEMS03	0.142	0.248	**0.124**	**0.231**	**12.7%**	0.239	0.329	**0.221**	**0.318**	**7.6%**	0.116	0.220	**0.112**	**0.216**	**3.4%**
PEMS04	0.121	0.232	**0.098**	**0.210**	**19.0%**	0.276	0.353	**0.258**	**0.339**	**6.5%**	0.120	0.222	**0.109**	**0.210**	**9.2%**
PEMS07	0.102	0.205	**0.082**	**0.183**	**19.6%**	0.210	0.305	**0.208**	**0.303**	**1.0%**	0.087	0.190	**0.080**	**0.181**	**8.0%**
PEMS08	0.254	0.306	**0.152**	**0.231**	**40.2%**	0.302	0.358	**0.271**	**0.336**	**9.6%**	0.148	0.213	**0.146**	**0.209**	**1.2%**
Exchange	0.368	0.409	**0.363**	**0.406**	**1.4%**	0.370	0.407	**0.365**	**0.404**	**1.4%**	0.425	0.448	**0.381**	**0.422**	**10.4%**
Weather	0.260	0.281	**0.250**	**0.275**	**3.8%**	0.240	0.262	**0.238**	**0.261**	**0.8%**	0.226	0.255	**0.220**	**0.252**	**2.7%**
Solar	0.234	0.261	**0.228**	**0.258**	**2.6%**	0.304	0.287	**0.296**	**0.281**	**2.6%**	0.211	0.243	**0.204**	**0.231**	**3.3%**
ECL	0.179	0.270	**0.168**	**0.262**	**6.1%**	0.177	0.263	**0.171**	**0.259**	**3.4%**	Out of Memory				
Traffic	0.428	0.282	**0.422**	**0.281**	**1.4%**	0.453	0.281	**0.436**	**0.268**	**3.8%**					
Average	0.279	0.315	**0.261**	**0.302**	**6.3%**	0.310	0.335	**0.301**	**0.329**	**3.0%**	0.243	0.298	**0.234**	**0.291**	**3.7%**
Best Count	2	2	**50**	**51**	-	9	11	**49**	**48**	-	2	7	**44**	**44**	-

Average 4Hs	PRformer MSE	PRformer MAE	+ CM MSE	+ CM MAE	Imp. (MSE)	Minusformer$^{L=96}$ MSE	Minusformer$^{L=96}$ MAE	+ CM MSE	+ CM MAE	Imp. (MSE)	Minusformer$^{L=336}$ MSE	Minusformer$^{L=336}$ MAE	+ CM MSE	+ CM MAE	Imp. (MSE)
ETTh1	0.444	0.445	**0.434**	**0.436**	**2.3%**	0.463	0.452	**0.451**	**0.446**	**2.6%**	0.453	0.453	**0.442**	**0.448**	**2.4%**
ETTh2	0.345	0.383	**0.340**	**0.381**	**1.4%**	0.394	0.409	**0.385**	**0.406**	**2.3%**	0.360	0.397	**0.356**	**0.395**	**1.1%**
ETTm1	0.352	0.376	**0.343**	**0.371**	**2.6%**	0.416	0.412	**0.404**	**0.408**	**1.9%**	0.369	0.393	**0.361**	**0.389**	**2.2%**
ETTm2	0.266	0.310	**0.262**	**0.317**	**1.5%**	0.285	0.328	**0.283**	**0.327**	**0.7%**	0.275	0.328	**0.263**	**0.322**	**4.4%**
PEMS03	0.152	0.254	**0.143**	**0.248**	**5.9%**	0.138	0.245	**0.114**	**0.223**	**17.4%**	0.087	0.190	**0.086**	**0.188**	**1.1%**
PEMS04	0.185	0.281	**0.173**	**0.272**	**6.5%**	0.171	0.270	**0.120**	**0.230**	**29.8%**	0.100	0.200	**0.093**	**0.195**	**7.0%**
PEMS07	0.140	0.239	**0.132**	**0.230**	**5.7%**	0.125	0.224	**0.089**	**0.191**	**28.8%**	0.068	0.165	**0.067**	**0.164**	**1.5%**
PEMS08	0.205	0.283	**0.186**	**0.271**	**9.3%**	0.170	0.254	**0.142**	**0.234**	**22.4%**	0.111	0.198	**0.109**	**0.191**	**1.8%**
Exchange	0.565	0.491	**0.472**	**0.449**	**16.5%**	0.508	0.488	**0.465**	**0.472**	**8.5%**	0.533	0.529	**0.474**	**0.500**	**11.1%**
Weather	0.224	0.256	**0.219**	**0.251**	**2.2%**	0.260	0.281	**0.252**	**0.275**	**3.1%**	0.242	0.275	**0.235**	**0.270**	**2.9%**
Solar	0.202	0.218	**0.197**	**0.212**	**2.5%**	0.230	0.253	**0.228**	**0.252**	**0.9%**	0.215	0.258	**0.212**	**0.255**	**1.4%**
ECL	0.155	0.246	**0.149**	**0.242**	**3.9%**	0.171	0.262	**0.164**	**0.258**	**4.1%**	0.161	0.254	**0.157**	**0.252**	**2.5%**
Traffic	0.378	0.236	**0.348**	**0.227**	**7.9%**	0.413	0.272	**0.405**	**0.268**	**1.9%**	0.373	0.260	**0.366**	**0.259**	**1.9%**
Average	0.278	0.309	**0.261**	**0.301**	**6.1%**	0.288	0.319	**0.269**	**0.307**	**6.6%**	0.258	0.300	**0.248**	**0.295**	**3.9%**
Best Count	3	4	51	49	-	4	8	**51**	**49**	-	6	8	**46**	**47**	-

Table 1: **Results of TS forecasting.** We apply our method to various backbones with varying input window sizes (L) and four horizons (H) over 13 datasets, yielding consistent performance gains.

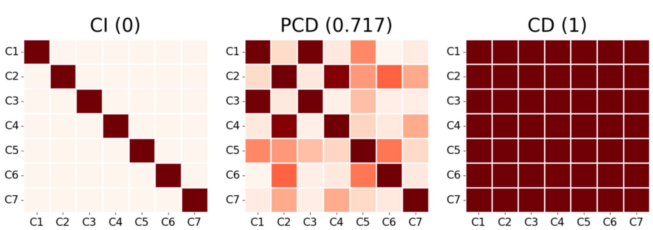

Fig. 6: CD ratio of CI, PCD, and CD.

with higher values indicating a greater preference for channel interactions. Note that it is a *relative* metric rather than an absolute one, as it is designed to compare preferences for CD across multiple datasets with a single model.

Figure 6 shows the visualization of $\mathbf{M}$ and its corresponding CD ratio for ETTh1 [20], with a ratio of 0.717 for PCD. We find that $\mathbf{M}$ effectively captures the degree of CD for each dataset, as datasets with higher $r(\mathbf{M})$ tend to have greater performance gains with CD architecture compared to CI architecture, as illustrated in Figure 7.

4 Experiments

We demonstrate the effectiveness of our method in both single-task models and TSFMs under supervised (SL) or self-supervised (SSL) settings, as:

- 1) Single-task model (SL): iTransformer [10], CARD [11], PRformer [18], MinusFormer [19]
- 2) Single-task model (SSL): TimeSiam [21] (discussed in Appendix E)
- 3) TSFM (SL, SSL): UniTS [22]

For UniTS, we consider four different tasks: forecasting (FCST), classification (CLS), imputation (IMP), and anomaly detection (AD), across various dataset sizes including few-shot and zero-shot settings. As evaluation metrics, we use the mean squared error (MSE) and mean absolute error (MAE) for FCST and IMP, accuracy (Acc.) for CLS, and F_1 score for AD. Dataset statistics and implementation details can be found in Appendix A and B, respectively.

[1] Forecasting & [2] Classification				
Ratio	Model		MSE	Acc.
5%	iTransformer	FT	0.598	51.4
	UniTS	PT	0.549	49.4
		FT	0.505	53.8
	UniTS + CM	PT	0.546	**54.9**
		FT	**0.489**	54.8
20%	iTransformer	FT	0.510	59.9
	UniTS	PT	0.525	58.9
		FT	0.486	63.6
	UniTS + CM	PT	0.453	60.0
		FT	**0.425**	**64.8**

[3] Imputation			
Ratio	Model		MSE
25%	iTransformer	FT	0.186
	UniTS	PT	0.179
		FT	0.167
	UniTS + CM	PT	0.179
		FT	**0.158**
50%	iTransformer	FT	0.226
	UniTS	PT	0.232
		FT	0.213
	UniTS + CM	PT	0.225
		FT	**0.201**

[4] Anomaly Detection		
Model		F_1
Anomaly Trans.	-	79.2
TimesNet	FT	74.2
PatchTST	FT	84.3
iTransformer	FT	83.1
UniTS	PT	81.7
	FT	85.6
UniTS + CM	PT	82.0
	FT	**86.6**

Table 2: **Few-shot.** The table presents the results of four few-shot tasks under both PT (prompt-tuning) and FT (fine-tuning) settings, reporting the average performance on [1] nine forecasting, [2] six classification, [3] six imputation, and [4] four anomaly detection tasks.

		UniTS	+ CM	Imp.
Forecasting (MSE)	Supervised	0.469	**0.445**	**5.1%**
	Prompt-tuning	0.478	**0.452**	**5.4%**
Classification (Acc.)	Supervised	80.6	**82.0**	**1.7%**
	Prompt-tuning	75.1	**78.3**	**4.3%**

Table 3: 20 FCST and 18 CLS tasks.

4.1 Application to Single-task Models

To demonstrate the effectiveness of CM, we apply it to four backbones[1] capturing CD with Transformer for forecasting tasks on 13 datasets. Table 1 presents the average performance across four horizons (H), demonstrating consistent gains across datasets and backbones with varying input window sizes (L) based on the original paper. Note that Minusformer selectively employs CI strategy for small datasets in its official code, whereas we use CD strategy for all datasets for fair comparisons. Additionally, we remove the dynamic projection module [23] from CARD, which generates summarized channel tokens, enabling application of CM without significantly impacting performance.

4.2 Application to TSFM

To validate the effectiveness of our method on a TSFM, we apply it to UniTS, which solves diverse tasks without the need for fine-tuning, relying solely on prompt-tuning.

a) Forecasting & classification. Table 3 presents a summary of the results from 20 forecasting tasks and 18 classification tasks under both supervised (Sup.) and prompt-tuning (PT) settings, with the full results provided in Appendix D.2 and Appendix D.1, respectively. The results indicate that applying our method enhances performance across both tasks. Notably, our method even outperforms single-task models that are individually trained for each task (dataset). Additionally, compared to GPT4TS [24], applying our method to UniTS achieves superior performance with less than 1% of the parameters (164.5M vs. 1.57M).

b) Few-shot learning. For the tasks under the few-shot settings, we conduct four different tasks (FCST, CLS, IMP, AD), following the experimental settings of UniTS. For (1) *FCST and CLS*, we experiment 9 FCST tasks and 6 CLS tasks with data ratios of 5% and 20%. For (2) *IMP*, we experiment 6 IMP tasks with a data ratio of 10%, where the goal is to impute 25% and 50% of missing data points. For (3) *AD*, we experiment 5 AD tasks with a data ratio of 5% Table 2 present the results, indicating that applying our method to UniTS yields the performance gain across all tasks, outperforming other single-task models [25, 7, 10].

c) Zero-shot learning. We perform FCST under two types of zero-shot settings: 1) *Zero-shot dataset*: We evaluate our model on an unseen dataset that was not included during training. 2) *Zero-shot task*: We assess the model's ability to predict a new forecasting horizon that was not encountered during training, by adding the mask tokens at the end of the TS to predict the desired future time steps.

For the FCST on unseen datasets, we evaluate our method using three datasets [26, 27, 28]. Table 4 presents the results, demonstrating consistent improvements by incorporating CMs. For the FCST with new forecasting horizons, we predict additional 384 time steps (by adding 24 masked tokens of length 16 at the end of the TS) on top of the base forecasting horizon of 96. Table 5 presents the results with four different datasets [20, 29], showing performance gains on three datasets.

[1] We use the official codes of each method for reproducibility.

Dataset	UniTS		+ CM		Imp.	
	MSE	MAE	MSE	MAE	MSE	MAE
Solar	0.597	0.607	**0.586**	**0.585**	**1.9%**	**3.6%**
River	**1.374**	0.698	**1.374**	**0.686**	0.0%	**1.7%**
Hospital	1.067	0.797	**1.020**	**0.777**	**4.4%**	**2.5%**
Avg.	1.013	0.701	**0.993**	**0.683**	**2.0%**	**2.6%**

Table 4: **Zero-shot.** Unseen datasets.

Dataset	UniTS		+ CM		Imp.	
	MSE	MAE	MSE	MAE	MSE	MAE
ECL	0.237	0.329	**0.231**	**0.323**	**2.5%**	**1.8%**
ETTh1	0.495	0.463	**0.492**	0.463	**0.6%**	0.0%
Traffic	0.632	0.372	**0.592**	**0.369**	**6.3%**	**0.8%**
Weather	0.335	0.336	0.335	0.336	0.0%	0.0%

Table 5: **Zero-shot.** New forecasting horizons.

CD by **model** ($\mathbf{QK}^\top/\sqrt{d_k}$)	CD by **dataset** ($\mathbf{M}$, ours)	Average MSE across four horizons													Avg.
		ETTh1	ETTh2	ETTm1	ETTm2	PEMS03	PEMS04	PEMS07	PEMS08	Exchange	Weather	Solar	ECL	Traffic	
✓		<u>0.457</u>	<u>0.384</u>	<u>0.408</u>	<u>0.293</u>	<u>0.142</u>	0.121	0.102	0.254	<u>0.368</u>	0.260	0.234	0.179	<u>0.428</u>	0.279
	✓	0.466	**0.383**	**0.398**	**0.289**	0.206	<u>0.116</u>	<u>0.101</u>	<u>0.162</u>	**0.363**	<u>0.259</u>	<u>0.233</u>	<u>0.176</u>	0.429	<u>0.275</u>
✓	✓	**0.444**	**0.383**	**0.398**	**0.289**	**0.124**	**0.098**	**0.082**	**0.152**	**0.363**	**0.250**	**0.228**	**0.168**	**0.422**	**0.261**

Table 6: **Effectiveness of CD by model and CD by dataset**. Using both the attention matrix (i.e., CD by model) and CM (i.e., CD by dataset) to capture local and global dependencies yields the best results, with CM alone outperforming the attention matrix in some cases.

Components		$\mathbf{A}$	FCST (20)		CLS (18)
Sim.	Dom.		MSE	MAE	Acc.
		$\mathbf{I}$	0.502	0.408	75.4%
		$\mathbf{1}$	0.478	0.393	75.1%
✓		$\lvert\mathbf{R}\rvert$	0.474	0.390	<u>78.8%</u>
✓		$\bar{\mathbf{R}}$	<u>0.471</u>	<u>0.388</u>	78.1%
	✓	$\sigma\left(\alpha \cdot \mathbf{I} + \beta\right)$	0.497	0.406	76.2%
✓	✓	$\sigma\left(\alpha \cdot \bar{\mathbf{R}} + \beta\right)$	**0.452**	**0.384**	**80.6%**

Table 7: Ablation study of CM.

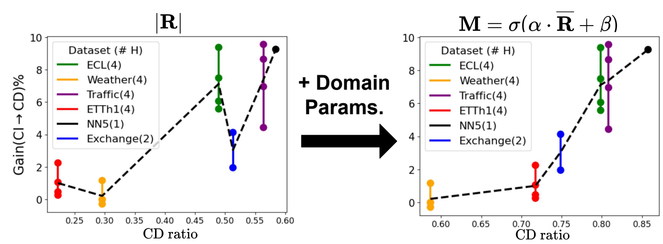

Fig. 7: CD ratio vs. Gain from (CI → CD).

4.3 Ablation Studies

Components of CM. To demonstrate the effectiveness of a CM, we conduct an ablation study of using the similarity matrix (Sim.) and the domain parameters (Dom.). Table 7 presents the result with 20 FCST and 18 CLS tasks with UniTS under the prompt-tuning setting, indicating that incorporating both components yields the best results. Note that, to isolate the effect of the domain parameters, we replace $\bar{\mathbf{R}}$ with an identity matrix ($\mathbf{I}$) in the fifth row of Table 7.

CD by model (attention matrix) & CD by dataset (CM). To demonstrate the importance of capturing both CD by model and CD by dataset, we conduct an ablation study, as shown in Table 6. Specifically, we use the attention weights $\mathbf{W}$ in $\mathrm{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{Softmax}\left(\mathbf{W}\right) \cdot \mathbf{V}$ in the following manner: 1) $\mathbf{QK}^\top/\sqrt{d_k}$ (i.e., attention matrix) for CD by model, 2) $\mathbf{M}$ (i.e., channel mask) for CD by dataset, and 3) $\mathbf{M} \odot \mathbf{QK}^\top/\sqrt{d_k}$ for both. The results show the average MSE across four horizons, indicating that using both components yields the best results. Notably, using only CMs provides better performance than using only attention matrices in some datasets.

5 Analysis

Effectiveness of domain parameters. To demonstrate the importance of domain parameters in reflecting the degree of CD, we compare 1) the CD ratio and 2) the performance gain achieved with the CD framework against the CI framework with UniTS. Figure 7 shows that the gain is highly correlated with the CD ratio of a CM with the domain parameters ($r(\mathbf{M})$), but less so without them ($r(|\mathbf{R}|)$).

CD ratio comparison. Table 8 presents the CD ratios of CMs with and without[2] domain parameters ($r(\mathbf{M})$ and $r(|\mathbf{R}|)$), when using UniTS. The results show that while datasets with higher $r(|\mathbf{R}|)$ generally have higher $r(\mathbf{M})$, this relationship is not consistent; for instance, Weather [29] exhibits lower CD despite having a stronger correlation compared to ETTh1 [20]. Figure 8 supports these findings by visualizing the channels of the datasets, revealing that the channels of ETTh1 tend to be more dependent on each other than those of Weather. These results underscore the importance of using domain parameters to adjust $|\mathbf{R}|$ for learning absolute dependencies specific to

[2]For a CM w/o domain parameters, we use the absolute correlation matrix ($|\mathbf{R}|$) instead of its mean-scaled version ($\bar{\mathbf{R}}$) to ensure a fair comparison with $\mathbf{M}$, which is also scaled between 0 and 1.

α, β	CD ratio	Weather	ILI	ETTh1	Exchange	ECL	Traffic	NN5
✗	$r(\lvert\mathbf{R}\rvert)$	0.296	0.708	0.222	0.513	0.489	0.564	0.584
✓	$r(\mathbf{M})$	0.587	0,706	0.717	0.749	0.800	0.808	0.857
		Low ⟵ $r(\mathbf{M})$ ⟶ High						
		Low ⟵ Dependencies btw channels ⟶ High						

Table 8: **CD ratio with and without domain parameters.** Applying domain parameters (α, β) aligns the CD ratio of CM ($r(\mathbf{M})$) with the dependencies between channels, which is as also supported in Figure 8.

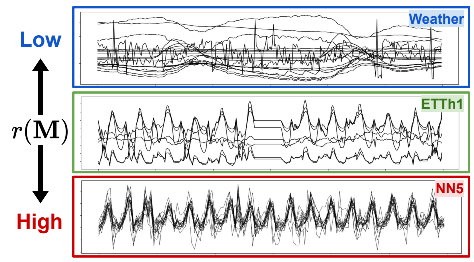

Fig. 8: TS visualization by $r(\mathbf{M})$.

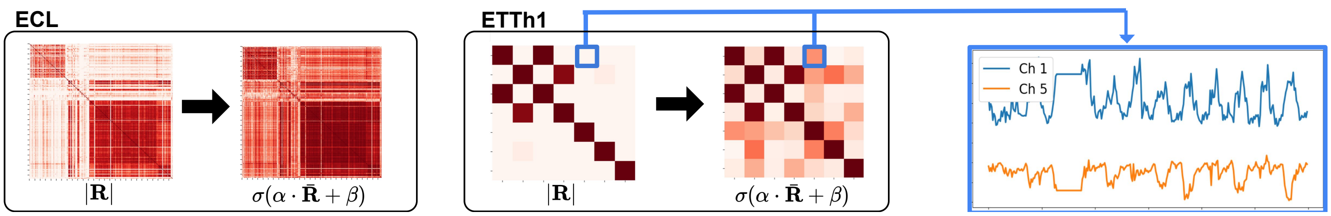

Fig. 9: **CM with and without domain parameters**. The figure shows CMs w/o domain parameters (e.g., similarity matrices) and CMs of two datasets, with each color scaled from 0 (light) to 1 (dark). The CM of ETTh1 reveals hidden relationships using domain parameters that are not captured by the correlation matrix.

Method	Dataset	MSE	MAE
UniTS		1.006	0.701
+ CM	Forecasting + Classification	0.995	0.684
	Forecasting	**0.993**	**0.683**
	Closest	**0.993**	**0.683**

Table 9: α, β for unseen datasets.

Dataset	w/o CM	w/ CM			
		Euc.	Cos.	DTW	Corr.
ETTh1	0.457	0.445	0.446	**0.444**	**0.444**
ETTh2	0.384	0.384	0.384	0.385	**0.383**
ETTm1	0.408	0.402	0.403	0.401	**0.398**
ETTm2	0.293	0.292	0.290	0.292	**0.289**
PEMS03	0.142	0.146	0.134	-	**0.124**
PEMS04	0.121	0.111	0.105	-	**0.098**
PEMS07	0.102	0.092	0.087	-	**0.082**
PEMS08	0.254	0.163	0.179	-	**0.152**
Exchange	0.368	0.364	**0.363**	0.364	**0.363**
Weather	0.260	0.256	0.255	0.254	**0.250**
Solar	0.234	0.232	0.229	-	**0.228**
ECL	0.179	0.173	0.171	-	**0.168**
Traffic	0.428	0.432	0.443	-	**0.422**
Avg.	0.279	0.269	0.268	-	**0.261**

Table 10: Various metrics for similarity.

each dataset. Furthermore, datasets with a larger number of channels tend to have higher $r(\mathbf{M})$, aligning with the prior work [30] emphasizing CD over CI for datasets with more channels.

Domain parameters for unseen dataset. For an unseen dataset, selecting the appropriate domain parameters is challenging, as these are not learned during training. To address this issue, we propose three strategies: 1) averaging the parameters across all datasets, 2) averaging the parameters from the forecasting datasets, and 3) selecting parameters from the dataset with the closest $r(\bar{\mathbf{R}})$, where Table 9 demonstrates the robustness of these strategies.

Visualization of CM. Figure 9 shows the CMs of ECL and ETTh1, illustrating the dependencies between the channels of each dataset. The CM of ETTh1 reveals a hidden relationship between the first and fifth channels when using domain parameters, which is not identified by $|\mathbf{R}|$ alone.

Various TS metrics. To demonstrate the effectiveness of CMs using metrics beyond (Pearson) correlation, we apply CMs to iTransformer with three different metrics: 1) Euclidean distance (Euc.), which we min-max normalize to the range (0,1) and subtract from 1 to convert it into a similarity metric; 2) cosine similarity (Cos.), for which we take the absolute value, following the same intuition as correlation; and 3) dynamic time warping (DTW), where we apply the same process as with Euc. Table 10 presents average MSE for $4H$s, indicating that CMs yield a performance gain regardless of the metric, with the best performance achieved with correlation. Note that we use DTW only for datasets with $C < 100$ due to its computational complexity.

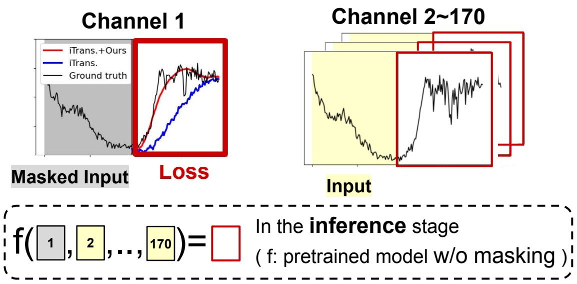

Fig. 10: Masked channel prediction.

H	PEMS04 ($C=307$)			PEMS08 ($C=170$)		
	iTrans.	+ CM	Imp.	iTrans.	+ CM	Imp.
12	0.549	**0.300**	**45.4%**	0.628	**0.200**	**68.1%**
24	0.718	**0.351**	**51.1%**	0.678	**0.241**	**64.5%**
48	0.750	**0.409**	**45.5%**	1.197	**1.059**	**11.5%**
96	0.758	**0.513**	**32.3%**	1.375	**1.217**	**11.5%**
Avg.	0.694	**0.393**	**43.3%**	0.970	**0.679**	**29.9%**

Table 11: Results of masked channel prediction.

$L, H = 96$	Weather ($C=21$)			ECL ($C=321$)		
	iTrans.	-	+ CM	iTrans.	-	+ CM
CD by model (Att. matrix)	✓		✓	✓		✓
CD by dataset (CM)		✓	✓		✓	✓
Train (sec/epoch)	26.2	24.1	26.7	33.2	26.0	36.4
Inference (ms)	11.1	11.1	11.2	12.4	11.0	13.2
Avg. MSE	0.260	0.259	**0.250**	0.179	0.176	**0.168**

Table 12: Efficiency analysis.

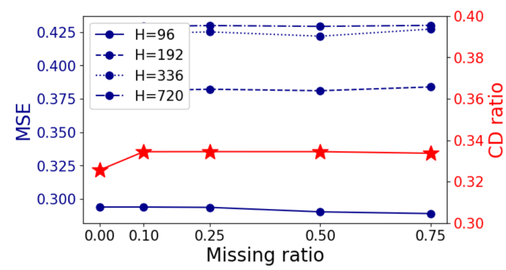

Fig. 11: Robustness to missingness.

Masked channel prediction. To evaluate the model's ability to capture CD, we introduce a novel evaluation method, *masked channel prediction*, which involves predicting the future values of the masked channel using the historical values of the unmasked channels. Specifically, we calculate the average loss for each channel when masked once, with the loss for the c-th channel expressed as:

$$L_{(c)}(y, \hat{y}) = \mathrm{MSE}(y[:, c], \hat{y}_{(c)}[:, c]), \text{where } \hat{y}_{(c)} = f(x_{(c)}), \tag{2}$$

where $x_{(c)}$ is x with the c-th channel masked, and $\hat{y}_{(c)}$ is the predicted output using $x_{(c)}$ as the input. Note that masked channel prediction is an *evaluation method* that does not require additional training, and instead uses a model pretrained without any masking.

To assess the effectiveness of CMs in capturing CD, we experiment masked channel prediction with iTransformer with and without CMs, imputing the historical values of the masked channels with there average values, which are essentially zero with normalization. The results in Table 11 demonstrate significant improvements by incorporating CMs. Furthermore, Figure 10 visualizes the predicted results for PEMS08 [31], where models with CMs predict masked channels better than models without CMs. We provide more results in Appendix I.

Efficiency analysis. To demonstrate the efficiency of CMs, we compare the training and inference times of iTransformer on two datasets [29] with varying numbers of channels, using 1) only attention matrices, 2) only CMs, and 3) both. Training time is measured per epoch and inference time is measured per data instance. Table 12 indicates that incorporating CMs *does not significantly impact computational time*, even with datasets containing a large number of channels. It is important to note that similarity matrices can be precomputed offline, making CMs practical for use.

Domain parameters		Channel mask ($\mathbf{M}$)	Asym.
Scalar	$\alpha, \beta \in \mathbb{R}^1$	$\sigma\left(\alpha \cdot \bar{\mathbf{R}} + \beta\right)$	✗
Vector	$\mathbf{E} \in \mathbb{R}^d$	$\mathrm{Norm}(\mathbf{E}\mathbf{E}^T) \odot \bar{\mathbf{R}}$	✗
	$\mathbf{E}_1, \mathbf{E}_2 \in \mathbb{R}^d$	$\mathrm{Norm}(\mathbf{E}_1\mathbf{E}_2^T) \odot \bar{\mathbf{R}}$	✓
Matrix	$\mathbf{A} \in \mathbb{R}^{C \times C}$	$\mathbf{A} \odot \bar{\mathbf{R}}$	✓

Table 13: Extension of domain parameters.

Robustness to missingness. To demonstrate the robustness of CMs to missing values, we analyze scenarios where some values are randomly missing at ratios of 10%, 25%, 50% and 75%, with the missing values linearly interpolated using adjacent values. Figure 11 shows the result on ETTh2 [20] with iTransformer, indicating that both $r(|\mathbf{R}|)$ and the performance remain robust to the missingness.

Various options for domain parameters. The proposed domain parameters (α and β) are scalars that adjust $\bar{\mathbf{R}}$ by changing its elements monotonically. For further flexibility, we design alternative options for the parameters: 1) a vector $\mathbf{E}$ for each channel and 2) a matrix $\mathbf{A}$ for each dataset. Both options are used to construct an adjustment matrix that is element-wise multiplied to $\bar{\mathbf{R}}$, as shown in Table 13.

The first option serves as identifiable vectors for each channel, with the adjustment matrix constructed based on the inner product between these vectors and normalized with $\mathrm{Norm}(\cdot) = \mathrm{Softmax}\left(\mathrm{ReLU}\left(\cdot\right)\right)$, while the second option

	Average MSE across four horizons													Avg.
	ETTh1	ETTh2	ETTm1	ETTm2	PEMS03	PEMS04	PEMS07	PEMS08	Exchange	Weather	Solar	ECL	Traffic	
α, β	**0.444**	**0.383**	**0.398**	**0.289**	**0.124**	**0.098**	**0.082**	**0.152**	**0.363**	**0.250**	<u>0.228</u>	**0.168**	0.422	**0.261**
$\mathbf{E}$	<u>0.452</u>	0.391	<u>0.402</u>	<u>0.291</u>	0.150	0.106	0.096	0.202	<u>0.364</u>	<u>0.255</u>	0.234	<u>0.177</u>	<u>0.416</u>	0.272
$\mathbf{E_1}, \mathbf{E_2}$	<u>0.452</u>	0.391	<u>0.402</u>	<u>0.291</u>	0.152	0.105	<u>0.095</u>	0.205	<u>0.364</u>	<u>0.255</u>	0.233	<u>0.177</u>	**0.415**	0.272
$\mathbf{A}$	0.454	0.391	<u>0.402</u>	<u>0.291</u>	<u>0.138</u>	<u>0.099</u>	0.102	<u>0.182</u>	<u>0.364</u>	0.259	**0.226**	<u>0.177</u>	0.418	<u>0.269</u>
-	0.457	<u>0.384</u>	0.408	0.293	0.142	0.121	0.102	0.254	0.368	0.260	0.234	0.179	0.428	0.279

Table 14: **Various domain parameters.** Using *scalar* parameters (α, β), which scale/shift the similarity matrix, yields the best results, compared to *vector* parameters ($\mathbf{E}$ or $\mathbf{E}_1, \mathbf{E}_2$) and *matrix* parameters ($\mathbf{A}$).

Univariate (e.g., flattening)	Multivariate	
	CI	CD
[33], [34], [35], [13]	[36], [37], [38]	[39], [40], [22], [41]

Table 15: Categorization of TSFMs.

acts as the adjustment matrix itself. For the vector parameters, we also implement an asymmetric matrix version that requires two different vectors for each channel: one for the inner vector ($\mathbf{E}_1$) and the other for the outer vector ($\mathbf{E}_2$), as described in the previous work [32]. Table 14 shows that using scalar parameters achieves the best performance, demonstrating the efficiency of CMs by requiring only two additional parameters per dataset.

6 Discussion

A. Scope of our work. This paper focuses on enhancing Transformer-based methods capturing CD, *rather than improving TSFMs*. We argue that *this focus is broadly impactful* for two reasons:

- **a) Transformer for CD.** Since the introduction of iTransformer [10], various methods have adopted complex attention mechanisms to capture CD, while simpler algorithms are used to capture TD; in this paper, we examine six representative methods [10, 19, 11, 18, 22, 21] that follow this framework.
- **b) Large-scale TS datasets.** The emergence of large-scale TS datasets further highlights the potential of our work, which emphasizes the importance of CD, as the capacity–robustness trade-off [9] indicates that large datasets tend to favor CD-based models, whereas smaller datasets benefit more from CI approaches.

Although we demonstrate applicability within UniTS [22], we leave the application to various TSFMs as future work due to its broader scope.

B. Application to other TSFMs. It is important to clarify that our focus is *not on improving TSFM*, but on enhancing general TS models that *capture CD with Transformer*. As shown in Table 15, various recent methods rely on attention-based approaches specifically to model CD.

Among these methods, we select UniTS [22] as the sole TSFM baseline in this study for the following reasons: 1) GTT [40] reformulates TS forecasting as an image task for next curve shape prediction and does not provide hyperparameter details or scripts for reproducibility. 2) UniST [41] is tailored for urban spatio-temporal prediction, taking spatial inputs in a 4D grid format which makes it incompatible with general TS benchmarks. 3) TTM [39] is built on TSMixer [16], which falls outside the scope of our work. In contrast, UniTS [22] handles four diverse tasks with a single architecture through prompt-tuning, making it a representative choice for demonstrating the effectiveness of CM.

Furthermore, Moirai [13], a CI method that applies time-axis attention after channel flattening, can still incorporate our CM by reshaping the CM from $C \times C$ to $C \cdot P \times C \cdot P$ with duplicated intra-channel values, where P is the number of patches, to match the shape of Moirai's attention matrix. Nonetheless, we exclude Moirai for the reasons detailed in Appendix F.

C. Effectiveness of CM under non-stationary TS.

- **a) Distribution shifts.** Adjusting the CM based on distribution shifts *would conflict with its intended purpose*, potentially diminishing its ability to capture stable, dataset-specific characteristics. Therefore, we rely on the attention matrix to capture these changes. Furthermore, the effectiveness of CM is evident under such shifts, as our datasets exhibit significant distribution shifts [9].
- **b) Time-lagged dependencies.** Since channels may exhibit time-lagged dependencies [3], we include metrics such as DTW in Table 10, which explicitly account for time lags between channels. Importantly, the potential challenge posed by lagged relationships pertains to the *choice of similarity metric*, not to the *use of similarity itself*, which is our primary focus. Nonetheless, as shown in Table 10, our method demonstrates robust performance regardless of the metric.

D. Comparison with other plug-in methods. The proposed CM is a *plug-in* method that can be applied orthogonally to other plug-in methods, including (Granger-causality based) LIFT [42] and PRReg [9]. Therefore, the key point is that *applying our method to various backbones consistently yields performance improvements*, rather than comparing with other plug-ins. While a comparison between these methods is still possible, their baseline models differ from ours, as our method employs more recent Transformer-based architectures, making a direct comparison infeasible. Furthermore, their code is not implemented as a plug-in method but is rather tightly coupled with backbones.

7 Conclusion

In this work, we introduce PCD, where CD captured by a Transformer-based model is adjusted using a CM, which is a plug-in method that leverages dataset-specific information to capture CD. Experimental results demonstrate that utilizing dataset-specific information is crucial for TS modeling, enhancing performance across various backbones. As our method is limited to Transformer-based models, we aim to develop a method to achieve PCD that *does not rely on the architecture*. We hope this work emphasizes the importance of utilizing dataset-specific information across diverse domains.

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Appendix

A Dataset Description

A.1 Dataset for Single-task Models

For TS forecasting in a single-task setting, we evaluate the effectiveness of our proposed method using 13 datasets, with their statistics described in Table A.1. We adhere to the same data processing and train-validation-test split protocol as iTransformer [10], ensuring that the training, validation, and test sets are separated in chronological order. The input length is consistently set to 96 across all datasets. Note that N and C denote the size of the dataset and number of channels in a dataset, respectively.

Dataset	C	Prediction Length	$(N_{\text{train}}, N_{\text{val}}, N_{\text{test}})$
ETTh1 [20]	7	{96, 192, 336, 720}	(8545, 2881, 2881)
ETTh2 [20]	7	{96, 192, 336, 720}	(8545, 2881, 2881)
ETTm1 [20]	7	{96, 192, 336, 720}	(34465, 11521, 11521)
ETTm2 [20]	7	{96, 192, 336, 720}	(34465, 11521, 11521)
Exchange [29]	8	{96, 192, 336, 720}	(5120, 665, 1422)
Weather [29]	21	{96, 192, 336, 720}	(36792, 5271, 10540)
ECL [29]	321	{96, 192, 336, 720}	(18317, 2633, 5261)
Traffic [29]	862	{96, 192, 336, 720}	(12185, 1757, 3509)
Solar-Energy [43]	137	{96, 192, 336, 720}	(36601, 5161, 10417)
PEMS03 [31]	358	{12, 24, 48, 96}	(15617, 5135, 5135)
PEMS04 [31]	307	{12, 24, 48, 96}	(10172, 3375, 3375)
PEMS07 [31]	883	{12, 24, 48, 96}	(16911, 5622, 5622)
PEMS08 [31]	170	{12, 24, 48, 96}	(10690, 3548, 3548)

Table A.1: Single-task forecasting datasets.

A.2 Dataset for Time Series Foundation Models

The datasets used in the experiment are aggregated from the Monash Forecasting Repository [44], the Time Series Classification Website [45], and the Time Series Library [25]. The combined training set includes more than 35 million time steps and over 6,000 variables (channels). Note that N, L, C denote the training size, input length, and number of channels in a dataset, respectively.

A.2.1 Multi-task Learning

For TS forecasting and classification in a multi-task setting, we evaluate the effectiveness of our proposed method using 20 datasets for forecasting and 18 datasets for classification. The statistics of these datasets are summarized in Table A.2 and A.3, respectively.

<table>
<tr><th>Category</th><th>Dataset</th><th>Prediction Length</th><th>N</th><th>L</th><th>C</th></tr>
<tr><td rowspan="2">Finance</td><td>NN5 [46]</td><td>112</td><td>409</td><td>112</td><td>111</td></tr>
<tr><td>Exchange [29]</td><td>192
336</td><td>5024
4880</td><td>96</td><td>8</td></tr>
<tr><td rowspan="2">Electricity</td><td>ECL [29]</td><td>96
192
336
720</td><td>18221
18125
17981
17597</td><td>96</td><td>321</td></tr>
<tr><td>ETTh1 [20]</td><td>96
192
336
720</td><td>8449
8353
8209
7825</td><td>96</td><td>7</td></tr>
<tr><td>Illness</td><td>ILI [29]</td><td>60</td><td>581</td><td>36</td><td>7</td></tr>
<tr><td>Traffic</td><td>Traffic [29]</td><td>96
192
336
720</td><td>12089
11993
11849
11465</td><td>96</td><td>862</td></tr>
<tr><td>Weather</td><td>Weather [29]</td><td>96
192
336
720</td><td>36696
36600
36456
36072</td><td>96</td><td>21</td></tr>
</table>

Table A.2: Multi-task forecasting datasets.

Category	Dataset	# classes	N	L	C
Finance	SharePriceIncrease [47]	2	965	60	1
Audio	JapaneseVowels [48]	9	270	29	12
	SpokenArabicDigits [48]	10	6599	93	13
	Heartbeat [48]	2	204	405	61
ECG	ECG5000 [47]	5	500	140	1
	NonInvasiveFetalECGThorax1 [47]	52	1800	750	1
EEG	Blink [48]	2	500	510	4
	FaceDetection [48]	2	5890	62	144
	SelfRegulationSCP2 [48]	2	200	1152	7
Sensors	ElectricDevices [47]	7	8926	96	1
	Trace [47]	4	100	275	1
	FordB [47]	2	3636	500	1
Human Activity	MotionSenseHAR [48]	6	966	200	12
	EMOPain [48]	3	968	180	30
	UWaveGestureLibrary [48]	8	120	315	3
Traffic	Chinatown [47]	2	20	24	1
	MelbournePedestrian [47]	10	1194	24	1
	PEMS-SF [48]	7	267	144	963

Table A.3: Multi-task classification datasets.

A.2.2 Few-shot Learning

For TS forecasting, classification, imputation, and anomaly detection in a few-shot setting, we evaluate the effectiveness of our proposed method using nine datasets for forecasting, six datasets for classification, four datasets for imputation, and five datasets for anomaly detection. The statistics of these datasets related to forecasting and classification are summarized in Table A.4, Table A.5, A.6, and A.7, respectively.

Category	Dataset	Prediction Length	N	L	C
Electricity	ETTh2 [20]	96	8449	96	7
		192	8353		
		336	8209		
		720	7825		
	ETTm1 [20]	96	34369	96	7
		192	34273		
		336	34129		
		720	33745		
Weather	SaugeenRiverFlow [27]	24	18921	48	1

Table A.4: Few-shot forecasting datasets.

Category	Dataset	# classes	N	L	C
ECG	ECG200 [47]	2	100	96	1
EEG	SelfRegulationSCP1 [48]	2	268	896	6
Human Activity	RacketSports [48]	4	151	30	6
	Handwriting [48]	26	150	152	3
	Epilepsy [48]	4	137	207	3
Sensor	StarLightCurves [47]	3	1000	1024	1

Table A.5: Few-shot classification datasets.

Category	Dataset	L	C
Electricity	ETTm1 [20]	96	7
	ETTh1 [20]	96	7
	ECL [29]	96	321
Weather	Weather [29]	96	21

Table A.6: Few-shot imputation datasets.

Category	Dataset	L	C
Machine	SMD [49]	96	38
	PSM [50]	96	25
Spacecraft	MSL [51]	96	55
	SMAP [51]	96	25
Infrastructure	SWaT [52]	96	51

Table A.7: Few-shot anomaly detection datasets.

A.2.3 Zero-shot Learning

For TS forecasting in a zero-shot setting, we evaluate the effectiveness of our proposed method using six datasets. Three of these datasets are used for the zero-shot setting with unseen datasets, while the remaining four datasets are used for the zero-shot setting with new prediction lengths. The statistics for the three unseen datasets are summarized in Table A.8.

Category	Dataset	Prediction Length	L	C
Electricity	Solar [26]	64	128	137
Weather	SaugeenRiverFlow [27]	128	256	1
Healthcare	Hospital [28]	16	32	767

Table A.8: Zero-shot forecasting datasets.

B Implementation Details

It is important to note that we follow the experimental settings of iTransformer for single-task and UniTS for multi-task settings, respectively. For the implementation, we use the official code repositories of both methods, running the provided scripts without modifications. However, for UniTS in the prompt tuning setting, we encountered an issue where the model failed to converge using the provided script. This was resolved by setting the hidden dimension to $D = 32$, which we applied uniformly across both UniTS and its integration with our method. The following sections outline the specific settings we adhered to.

B.1 Implementation for Single-task Models

iTransformer. Following iTransformer [10], we use the Adam optimizer [53] and L2 loss for model optimization. The batch size is consistently set to 32, and the number of training epochs is fixed at 10. Since our approach is plug-and-play, we do not adjust any hyperparameters for our method; instead, we use the same hyperparameters employed by iTransformer. For all other single-task models, including CARD [11], Minusformer [19], and PRformer [18], we follow the experimental setups provided in their original works.

B.2 Implementation for Time Series Foundation Models

Model architecture. In a multi-task setting, the UniTS network consists of three UniTS blocks, along with one `GEN` tower and one `CLS` tower. For each data source, specific prompt and task tokens are assigned, with forecasting tasks on the same source but with varying forecast lengths using the same prompt and `GEN` token. To enable zero-shot learning on new datasets, a shared prompt and `GEN` token are applied across all data sources. The embedding dimensions are set to 64 for the supervised version, and 32 for the prompt-tuning version, and all blocks in UniTS retain the same feature shape.

Model training. In multi-task settings, models are trained jointly on multiple tasks following a unified protocol. To match the largest dataset, samples from each dataset are repeated within each epoch. Supervised training is conducted over 5 epochs with gradient accumulation, yielding an effective batch size of 1024. The initial learning rate is set at 3.2e-2 and is adjusted using a multi-step decay schedule. For self-supervised pretraining, the models training with an are trained for 10 epochs with effective batch size of 4096, starting with a learning rate of 6.4e-3, which is adjusted using a cosine decay schedule.

B.3 Construction of Correlation Matrix

For constructing the correlation matrix for CM, we used the datasets corresponding to the training period for forecasting datasets and the training instances for classification datasets. Specifically, for a forecasting dataset with shape $(C, L_{\text{train}} + L_{\text{val}} + L_{\text{test}})$, we compute the correlation matrix with shape (C, C) using only the training period with shape (C, L_{train}). For a classification dataset with shape $(N_{\text{train}} + N_{\text{val}} + N_{\text{test}}, C, L)$, we compute the correlation matrix with shape (C, C) using only the training instances with shape (N_{train}, C, L) by averaging across the instances.

C Application to Single-task Models

To demonstrate the effectiveness of our method on a model with a single-task setting, we apply it to the TS forecasting task using four backbones on 13 datasets.

Metric		iTransformer		+ CM	
		MSE	MAE	MSE	MAE
ETTh1	96	0.387	0.405	**0.385**	**0.404**
	192	0.441	0.436	**0.438**	**0.434**
	336	0.491	0.462	**0.475**	**0.454**
	720	0.509	0.494	**0.477**	**0.474**
	Avg.	0.457	0.449	**0.444**	**0.441**
ETTh2	96	0.301	0.350	**0.295**	**0.347**
	192	0.381	0.399	**0.380**	**0.397**
	336	**0.423**	**0.432**	0.427	0.434
	720	**0.430**	0.446	0.432	**0.445**
	Avg.	0.384	0.407	**0.383**	**0.406**
ETTm1	96	0.342	0.377	**0.331**	**0.369**
	192	0.383	0.396	**0.372**	**0.390**
	336	0.418	0.418	**0.412**	**0.414**
	720	0.487	0.456	**0.479**	**0.453**
	Avg.	0.408	0.412	**0.398**	**0.406**
ETTm2	96	0.186	**0.272**	**0.184**	**0.272**
	192	0.254	0.314	**0.251**	**0.311**
	336	0.317	0.353	**0.312**	**0.350**
	720	0.416	0.409	**0.412**	**0.408**
	Avg.	0.293	0.337	**0.289**	**0.335**
Exchange	96	0.086	0.206	**0.085**	**0.205**
	192	0.181	0.303	**0.180**	**0.302**
	336	0.338	0.422	**0.337**	**0.421**
	720	0.869	0.704	**0.850**	**0.696**
	Avg.	0.368	0.409	**0.363**	**0.406**
Weather	96	0.174	0.215	**0.165**	**0.209**
	192	0.224	0.258	**0.213**	**0.251**
	336	0.281	0.298	**0.274**	**0.296**
	720	0.359	0.351	**0.350**	**0.346**
	Avg.	0.260	0.281	**0.250**	**0.275**
Solar	96	0.201	0.234	**0.197**	**0.231**
	192	0.238	0.263	**0.232**	**0.260**
	336	0.248	0.273	**0.241**	**0.270**
	720	0.249	0.275	**0.241**	**0.273**
	Avg.	0.234	0.261	**0.228**	**0.258**

Metric		iTransformer		+ CM	
		MSE	MAE	MSE	MAE
PEMS03	12	0.071	0.174	**0.063**	**0.168**
	24	0.097	0.208	**0.087**	**0.197**
	48	0.161	0.272	**0.133**	**0.250**
	96	0.240	0.338	**0.212**	**0.316**
	Avg.	0.142	0.248	**0.124**	**0.231**
PEMS04	12	0.081	0.188	**0.075**	**0.181**
	24	0.099	0.211	**0.086**	**0.196**
	48	0.133	0.246	**0.108**	**0.222**
	96	0.172	0.283	**0.125**	**0.242**
	Avg.	0.121	0.232	**0.098**	**0.210**
PEMS07	12	0.067	0.165	**0.061**	**0.157**
	24	0.088	0.190	**0.076**	**0.179**
	48	0.113	0.218	**0.086**	**0.188**
	96	0.140	0.246	**0.104**	**0.208**
	Avg.	0.102	0.205	**0.082**	**0.183**
PEMS08	12	0.088	0.193	**0.085**	**0.190**
	24	0.138	0.243	**0.126**	**0.234**
	48	0.334	0.353	**0.178**	**0.241**
	96	0.458	0.436	**0.221**	**0.260**
	Avg.	0.254	0.306	**0.152**	**0.231**
ECL	96	0.148	0.240	**0.140**	**0.235**
	192	0.167	0.258	**0.158**	**0.252**
	336	0.179	0.272	**0.172**	**0.267**
	720	0.220	0.310	**0.202**	**0.295**
	Avg.	0.179	0.270	**0.168**	**0.262**
Traffic	96	0.395	0.268	**0.391**	**0.266**
	192	0.417	0.277	**0.409**	**0.275**
	336	0.433	0.283	**0.426**	**0.282**
	720	0.467	0.300	**0.460**	**0.300**
	Avg.	0.428	0.282	**0.422**	**0.281**
Best Count (/52)		2	2	50	51

Table C.1: TS forecasting results with **iTransformer** with $L = 96$.

Metric		CARD MSE	CARD MAE	+ CM MSE	+ CM MAE
ETTh1	96	0.383	0.391	**0.378**	**0.386**
	192	0.437	0.420	**0.431**	**0.417**
	336	0.477	0.440	**0.472**	**0.438**
	720	0.477	0.463	**0.471**	**0.459**
	Avg.	0.444	0.429	**0.438**	**0.425**
ETTh2	96	0.281	**0.329**	**0.279**	**0.329**
	192	0.359	0.381	**0.355**	**0.377**
	336	**0.395**	**0.410**	0.403	0.415
	720	**0.403**	0.427	**0.403**	**0.426**
	Avg.	0.360	0.387	**0.360**	**0.387**
ETTm1	96	0.314	0.345	**0.312**	**0.343**
	192	0.361	**0.368**	**0.360**	**0.368**
	336	0.394	0.390	**0.389**	**0.388**
	720	0.463	0.426	**0.455**	**0.425**
	Avg.	0.383	0.382	**0.379**	**0.381**
ETTm2	96	0.170	0.248	**0.167**	**0.246**
	192	0.234	0.291	**0.231**	**0.289**
	336	0.293	**0.328**	**0.291**	**0.328**
	720	**0.391**	**0.388**	**0.391**	**0.388**
	Avg.	0.272	0.314	**0.270**	**0.313**
Exchange	96	0.083	0.200	**0.082**	**0.199**
	192	0.178	0.298	**0.175**	**0.297**
	336	0.350	0.425	**0.336**	**0.417**
	720	0.870	**0.703**	**0.869**	0.704
	Avg.	0.370	0.407	**0.365**	**0.404**
Weather	96	0.152	0.190	**0.150**	**0.188**
	192	0.203	0.239	**0.200**	**0.236**
	336	0.261	**0.283**	**0.260**	**0.283**
	720	0.343	0.337	**0.342**	**0.336**
	Avg.	0.240	0.262	**0.238**	**0.261**
Solar	96	**0.244**	**0.254**	0.248	0.255
	192	0.305	0.288	**0.289**	**0.276**
	336	0.347	0.307	**0.318**	**0.296**
	720	**0.318**	0.298	0.328	**0.297**
	Avg.	0.304	0.287	**0.296**	**0.281**

Metric		CARD MSE	CARD MAE	+ CM MSE	+ CM MAE
PEMS03	12	0.091	0.202	**0.089**	**0.201**
	24	**0.139**	**0.251**	**0.139**	0.253
	48	0.244	0.345	**0.238**	**0.338**
	96	0.480	0.517	**0.420**	**0.478**
	Avg.	0.239	0.329	**0.221**	**0.318**
PEMS04	12	**0.107**	0.218	**0.107**	**0.216**
	24	0.173	0.278	**0.166**	**0.274**
	48	0.309	0.380	**0.285**	**0.363**
	96	0.514	0.534	**0.473**	**0.502**
	Avg.	0.276	0.353	**0.258**	**0.339**
PEMS07	12	0.080	0.187	**0.078**	**0.185**
	24	**0.134**	**0.241**	**0.134**	**0.241**
	48	0.240	0.339	**0.239**	**0.338**
	96	0.387	0.451	**0.383**	**0.448**
	Avg.	0.210	0.305	**0.208**	**0.303**
PEMS08	12	0.103	0.210	**0.099**	**0.205**
	24	0.165	0.270	**0.159**	**0.263**
	48	0.311	0.390	**0.281**	**0.359**
	96	0.628	0.561	**0.545**	**0.517**
	Avg.	0.302	0.358	**0.271**	**0.336**
ECL	96	0.146	0.236	**0.145**	**0.235**
	192	**0.165**	**0.248**	**0.165**	**0.248**
	336	0.179	0.270	**0.174**	**0.264**
	720	0.216	0.299	**0.205**	**0.290**
	Avg.	0.177	0.263	**0.171**	**0.259**
Traffic	96	0.419	0.269	**0.402**	**0.254**
	192	0.443	0.275	**0.424**	**0.263**
	336	0.459	0.283	**0.444**	**0.270**
	720	0.489	0.298	**0.473**	**0.286**
	Avg.	0.453	0.281	**0.436**	**0.268**
Best Count (/52)		9	11	**49**	**48**

Table C.2: TS forecasting results with **CARD** with $L = 96$.

Metric		CARD MSE	CARD MAE	+ CM MSE	+ CM MAE
ETTh1	96	0.372	**0.400**	**0.370**	**0.400**
	192	**0.408**	**0.425**	**0.408**	**0.424**
	336	0.418	0.431	**0.417**	**0.430**
	720	0.427	0.457	**0.424**	**0.455**
	Avg.	0.406	0.428	**0.405**	**0.427**
ETTh2	96	0.269	0.330	**0.267**	**0.329**
	192	0.338	0.376	**0.335**	**0.373**
	336	0.330	**0.380**	**0.328**	**0.380**
	720	0.384	**0.424**	**0.382**	**0.424**
	Avg.	0.330	0.378	**0.328**	**0.377**
ETTm1	96	0.288	**0.331**	**0.285**	**0.331**
	192	0.330	0.357	**0.328**	**0.356**
	336	0.364	0.379	**0.362**	**0.377**
	720	0.418	**0.411**	**0.415**	**0.411**
	Avg.	0.350	0.370	**0.347**	**0.369**
ETTm2	96	0.161	0.247	**0.160**	**0.246**
	192	0.217	0.287	**0.215**	**0.285**
	336	0.269	0.322	**0.266**	**0.320**
	720	0.362	0.381	**0.359**	**0.380**
	Avg.	0.252	0.309	**0.250**	**0.308**
Exchange	96	0.104	0.228	**0.100**	**0.224**
	192	0.255	0.365	**0.246**	**0.357**
	336	0.405	0.464	**0.381**	**0.449**
	720	0.937	0.735	**0.796**	**0.657**
	Avg.	0.425	0.448	**0.381**	**0.422**
Weather	96	0.145	0.188	**0.144**	**0.187**
	192	0.189	0.230	**0.188**	**0.229**
	336	0.243	0.274	**0.240**	**0.271**
	720	0.316	0.329	**0.308**	**0.322**
	Avg.	0.226	0.255	**0.220**	**0.252**
Solar	96	0.171	0.211	**0.164**	**0.210**
	192	0.210	0.234	**0.201**	**0.229**
	336	0.227	0.246	**0.225**	**0.242**
	720	0.235	0.280	**0.228**	**0.244**
	Avg.	0.211	0.243	**0.204**	**0.231**

Metric		CARD MSE	CARD MAE	+ CM MSE	+ CM MAE
PEMS03	12	**0.064**	**0.168**	**0.064**	**0.168**
	24	0.095	0.202	**0.083**	**0.188**
	48	0.145	0.251	**0.139**	**0.247**
	96	0.167	0.264	**0.163**	**0.261**
	Avg.	0.116	0.220	**0.112**	**0.216**
PEMS04	12	0.081	0.185	**0.079**	**0.181**
	24	0.100	0.205	**0.098**	**0.201**
	48	0.152	0.250	**0.122**	**0.224**
	96	0.146	0.245	**0.137**	**0.235**
	Avg.	0.120	0.222	**0.109**	**0.210**
PEMS07	12	0.060	0.160	**0.055**	**0.154**
	24	0.079	0.184	**0.075**	**0.180**
	48	0.094	0.197	**0.088**	**0.189**
	96	0.112	0.219	**0.103**	**0.202**
	Avg.	0.087	0.190	**0.080**	**0.181**
PEMS08	12	0.085	0.180	**0.080**	**0.177**
	24	0.111	0.199	**0.109**	**0.198**
	48	0.167	0.228	**0.160**	**0.224**
	96	0.241	0.246	**0.235**	**0.238**
	Avg.	0.148	0.213	**0.146**	**0.209**
Best Count (/44)		2	7	44	44

Table C.3: TS forecasting results with **CARD** with $L = 720$.

Metric		PRformer MSE	PRformer MAE	+ CM MSE	+ CM MAE
ETTh1	96	0.361	0.386	**0.360**	**0.384**
	192	**0.409**	**0.416**	**0.409**	0.419
	336	0.467	0.457	**0.456**	**0.446**
	720	0.540	0.522	**0.508**	**0.497**
	Avg.	0.444	0.445	**0.434**	**0.436**
ETTh2	96	0.284	0.335	**0.278**	**0.332**
	192	0.336	**0.374**	**0.334**	0.378
	336	0.363	0.398	**0.358**	**0.395**
	720	0.398	0.424	**0.389**	**0.420**
	Avg.	0.345	0.383	**0.340**	**0.381**
ETTm1	96	0.282	0.335	**0.276**	**0.330**
	192	0.332	0.363	**0.325**	**0.360**
	336	0.365	0.385	**0.359**	**0.381**
	720	0.427	0.421	**0.414**	**0.414**
	Avg.	0.352	0.376	**0.343**	**0.371**
ETTm2	96	**0.164**	**0.248**	**0.164**	**0.248**
	192	**0.225**	**0.292**	0.230	0.295
	336	0.287	0.336	**0.285**	**0.335**
	720	0.387	0.399	**0.374**	**0.393**
	Avg.	0.266	0.319	**0.262**	**0.317**
Exchange	96	0.093	0.215	**0.084**	**0.204**
	192	0.211	0.328	**0.181**	**0.304**
	336	0.396	0.468	**0.371**	**0.449**
	720	1.558	0.952	**1.251**	**0.840**
	Avg.	0.565	0.491	**0.472**	**0.449**
Weather	96	0.146	0.188	**0.143**	**0.183**
	192	0.192	0.233	**0.189**	**0.230**
	336	0.245	0.277	**0.239**	**0.271**
	720	0.310	0.324	**0.306**	**0.321**
	Avg.	0.224	0.256	**0.219**	**0.251**
Solar	96	0.170	0.199	**0.167**	**0.195**
	192	0.193	0.213	**0.188**	**0.209**
	336	0.217	0.227	**0.211**	**0.220**
	720	0.229	0.234	**0.222**	**0.226**
	Avg.	0.202	0.218	**0.197**	**0.212**

Metric		PRformer MSE	PRformer MAE	+ CM MSE	+ CM MAE
PEMS03	12	0.077	0.182	**0.072**	**0.177**
	24	0.106	0.215	**0.103**	**0.211**
	48	0.164	0.271	**0.161**	**0.269**
	96	0.259	0.346	**0.238**	**0.337**
	Avg.	0.152	0.254	**0.143**	**0.248**
PEMS04	12	0.094	0.199	**0.088**	**0.193**
	24	0.135	0.240	**0.123**	**0.231**
	48	0.213	0.306	**0.190**	**0.292**
	96	0.300	0.379	**0.292**	**0.373**
	Avg.	0.185	0.281	**0.173**	**0.272**
PEMS07	12	0.071	0.167	**0.064**	**0.157**
	24	0.100	0.201	**0.093**	**0.195**
	48	0.153	0.255	**0.145**	**0.249**
	96	0.235	0.331	**0.225**	**0.318**
	Avg.	0.140	0.239	**0.132**	**0.230**
PEMS08	12	0.087	0.189	**0.083**	**0.185**
	24	0.127	0.229	**0.119**	**0.222**
	48	0.211	0.301	**0.193**	**0.287**
	96	0.394	0.414	**0.349**	**0.390**
	Avg.	0.205	0.283	**0.186**	**0.271**
ECL	96	0.127	0.216	**0.124**	**0.214**
	192	0.148	0.237	**0.144**	**0.234**
	336	0.161	0.254	**0.157**	**0.250**
	720	0.185	0.278	**0.172**	**0.269**
	Avg.	0.155	0.246	**0.149**	**0.242**
Traffic	96	0.349	0.222	**0.326**	**0.214**
	192	0.357	0.225	**0.332**	**0.220**
	336	0.385	0.239	**0.348**	**0.228**
	720	0.421	0.258	**0.388**	**0.245**
	Avg.	0.378	0.236	**0.348**	**0.227**
Best Count (/52)		3	4	**51**	**49**

Table C.4: TS forecasting results with **PRformer** with varying L by datasets.

Metric		Minusformer		+ CM	
		MSE	MAE	MSE	MAE
ETTh1	96	**0.387**	**0.404**	**0.387**	**0.404**
	192	**0.446**	0.437	**0.446**	**0.436**
	336	0.502	0.473	**0.488**	**0.464**
	720	0.517	0.494	**0.485**	**0.481**
	Avg.	0.463	0.452	**0.451**	**0.446**
ETTh2	96	0.310	**0.352**	**0.306**	**0.352**
	192	0.390	0.403	**0.379**	**0.397**
	336	0.454	0.443	**0.438**	**0.437**
	720	0.420	**0.437**	**0.417**	**0.437**
	Avg.	0.394	0.409	**0.385**	**0.406**
ETTm1	96	0.341	0.371	**0.325**	**0.362**
	192	0.380	0.390	**0.370**	**0.385**
	336	**0.437**	**0.424**	0.441	0.430
	720	0.507	0.462	**0.479**	**0.454**
	Avg.	0.416	0.412	**0.404**	**0.408**
ETTm2	96	0.180	**0.260**	**0.179**	**0.260**
	192	0.243	**0.303**	**0.242**	**0.303**
	336	0.309	0.345	**0.304**	**0.343**
	720	0.407	0.403	**0.406**	**0.403**
	Avg.	0.285	0.328	**0.283**	**0.327**
Exchange	96	**0.096**	**0.227**	**0.096**	0.229
	192	0.222	0.353	**0.220**	**0.351**
	336	0.463	0.516	**0.416**	**0.487**
	720	1.251	0.856	**1.130**	**0.824**
	Avg.	0.508	0.488	**0.465**	**0.472**
Weather	96	0.174	0.213	**0.166**	**0.207**
	192	0.228	0.261	**0.217**	**0.253**
	336	0.281	0.299	**0.274**	**0.295**
	720	0.357	0.349	**0.351**	**0.346**
	Avg.	0.260	0.281	**0.252**	**0.275**
Solar	96	0.202	**0.231**	**0.200**	**0.231**
	192	**0.227**	**0.248**	0.229	0.249
	336	0.245	0.264	**0.239**	**0.262**
	720	0.246	0.267	**0.243**	**0.266**
	Avg.	0.230	0.253	**0.228**	**0.252**

Metric		Minusformer		+ CM	
		MSE	MAE	MSE	MAE
PEMS03	12	0.067	0.173	**0.063**	**0.167**
	24	0.095	0.206	**0.085**	**0.193**
	48	0.149	0.261	**0.123**	**0.236**
	96	0.239	0.341	**0.185**	**0.297**
	Avg.	0.138	0.245	**0.114**	**0.223**
PEMS04	12	0.085	0.190	**0.076**	**0.180**
	24	0.118	0.226	**0.095**	**0.205**
	48	0.179	0.282	**0.127**	**0.241**
	96	0.303	0.382	**0.181**	**0.296**
	Avg.	0.171	0.270	**0.120**	**0.230**
PEMS07	12	0.063	0.160	**0.057**	**0.152**
	24	0.090	0.192	**0.074**	**0.175**
	48	0.137	0.238	**0.095**	**0.200**
	96	0.208	0.304	**0.130**	**0.237**
	Avg.	0.125	0.224	**0.089**	**0.191**
PEMS08	12	0.077	0.177	**0.074**	**0.174**
	24	0.113	0.213	**0.101**	**0.203**
	48	0.182	0.272	**0.147**	**0.242**
	96	0.308	0.354	**0.245**	**0.318**
	Avg.	0.170	0.254	**0.142**	**0.234**
ECL	96	0.144	0.236	**0.136**	**0.231**
	192	0.161	0.252	**0.154**	**0.248**
	336	0.174	0.267	**0.169**	**0.265**
	720	0.204	0.294	**0.196**	**0.290**
	Avg.	0.171	0.262	**0.164**	**0.258**
Traffic	96	0.400	0.267	**0.393**	**0.264**
	192	0.400	0.264	**0.397**	**0.262**
	336	0.416	0.271	**0.406**	**0.267**
	720	0.434	0.284	**0.425**	**0.280**
	Avg.	0.413	0.272	**0.405**	**0.268**
Best Count (/52)		4	8	51	49

Table C.5: TS forecasting results with **Minusformer** with $L = 96$.

Metric		Minusformer		+ CM	
		MSE	MAE	MSE	MAE
ETTh1	96	0.384	0.406	**0.381**	**0.404**
	192	0.431	0.432	**0.424**	**0.427**
	336	0.527	0.498	**0.514**	**0.491**
	720	0.469	0.476	**0.451**	**0.470**
	Avg.	0.453	0.453	**0.442**	**0.448**
ETTh2	96	0.289	0.348	**0.287**	**0.347**
	192	0.356	0.388	**0.349**	**0.386**
	336	0.378	0.411	**0.374**	**0.408**
	720	0.418	0.442	**0.413**	**0.439**
	Avg.	0.360	0.397	**0.356**	**0.395**
ETTm1	96	0.302	0.352	**0.296**	**0.348**
	192	0.346	0.380	**0.344**	**0.378**
	336	0.379	0.397	**0.369**	**0.392**
	720	0.449	0.444	**0.437**	**0.437**
	Avg.	0.369	0.393	**0.361**	**0.389**
ETTm2	96	0.175	0.262	**0.173**	**0.261**
	192	0.237	0.310	**0.226**	**0.301**
	336	0.286	0.340	**0.279**	**0.335**
	720	0.400	0.401	**0.376**	**0.391**
	Avg.	0.275	0.328	**0.263**	**0.322**
Exchange	96	0.157	0.297	**0.136**	**0.273**
	192	**0.291**	**0.417**	0.305	0.427
	336	0.560	0.582	**0.460**	**0.516**
	720	1.125	0.821	**0.997**	**0.784**
	Avg.	0.533	0.529	**0.474**	**0.500**
Weather	96	0.161	0.210	**0.154**	**0.205**
	192	0.206	0.250	**0.199**	**0.247**
	336	0.256	0.291	**0.251**	**0.285**
	720	0.346	0.348	**0.337**	**0.344**
	Avg.	0.242	0.275	**0.235**	**0.270**
Solar	96	**0.216**	**0.238**	0.218	0.240
	192	0.206	0.255	**0.198**	**0.249**
	336	0.218	0.268	**0.216**	**0.265**
	720	0.220	0.269	**0.218**	**0.266**
	Avg.	0.215	0.258	**0.212**	**0.255**

Metric		Minusformer		+ CM	
		MSE	MAE	MSE	MAE
PEMS03	12	0.059	0.158	**0.056**	**0.156**
	24	0.071	0.174	**0.070**	**0.172**
	48	0.104	0.213	**0.098**	**0.204**
	96	**0.114**	**0.216**	0.120	0.220
	Avg.	0.087	0.190	**0.086**	**0.188**
PEMS04	12	0.072	0.171	**0.070**	**0.169**
	24	0.085	0.186	**0.080**	**0.182**
	48	0.116	0.222	**0.109**	**0.218**
	96	0.125	0.221	**0.114**	**0.214**
	Avg.	0.100	0.200	**0.093**	**0.195**
PEMS07	12	0.052	0.143	**0.050**	**0.142**
	24	0.061	0.154	**0.058**	**0.153**
	48	0.083	**0.184**	**0.080**	**0.184**
	96	**0.076**	**0.179**	0.079	**0.179**
	Avg.	0.068	0.165	**0.067**	**0.164**
PEMS08	12	0.066	0.170	**0.064**	**0.162**
	24	0.083	**0.175**	**0.080**	0.176
	48	0.137	0.221	**0.132**	**0.212**
	96	**0.157**	**0.213**	0.159	0.216
	Avg.	0.111	0.198	**0.109**	**0.191**
ECL	96	0.131	0.224	**0.127**	**0.223**
	192	0.153	0.246	**0.152**	**0.245**
	336	0.166	0.260	**0.162**	**0.258**
	720	0.194	0.287	**0.188**	**0.261**
	Avg.	0.161	0.254	**0.157**	**0.252**
Traffic	96	0.350	0.250	**0.338**	**0.246**
	192	0.377	**0.261**	**0.376**	**0.261**
	336	0.377	0.261	**0.361**	**0.257**
	720	**0.386**	0.268	0.389	**0.269**
	Avg.	0.373	0.260	**0.366**	**0.259**
Best Count (/52)		6	8	46	47

Table C.6: TS forecasting results with **Minusformer** with $L = 336$.

D Application to UniTS

To demonstrate the effectiveness of our method on a TS foundation model, we apply it to four different TS tasks using UniTS [22] on datasets from various domains, under multiple settings, including multi-task, few-shot, and zero-shot settings. All experimental settings follow those outlined in UniTS [22]. The sections and tables outlining the full experiment results are listed in Table D.1.

Settings	Section	TS downstream tasks			
		FCST	CLS	IMP	AD
Multi-task	D.1	Table D.2	Table D.3	-	-
Few-shot	D.2	Table D.4,D.5,D.6	Table D.7,D.8,D.9	Table D.10	Table D.11
Zero-shot	4.2	Table 4, 5	-	-	-

Table D.1: Summary of experiments.

D.1 Multi-task Learning

For experiments under multi-task settings, we perform 20 TS forecasting and 18 classification tasks, where the full results are shown in Table D.2 and Table D.3, respectively.

20 Tasks		Shared (1 model)								Task-specific (20 models)							
		UniTS + CM				UniTS				iTransformer		TimesNet		PatchTST		GPT4TS	
		Sup.		PT		Sup.		PT		Sup.						FT	
Dataset	H	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
NN5	112	0.641	0.568	**0.586**	**0.536**	0.635	0.556	0.611	0.552	0.623	0.554	0.629	0.541	0.634	0.568	0.623	0.545
ECL	96	0.176	0.278	**0.168**	**0.272**	0.172	0.273	0.174	0.277	0.204	0.288	0.184	0.289	0.212	0.299	0.198	0.285
	192	0.188	0.287	**0.184**	0.286	0.185	**0.284**	0.189	0.289	0.208	0.294	0.204	0.307	0.213	0.303	0.200	0.288
	336	0.199	**0.295**	0.199	0.301	**0.196**	0.297	0.205	0.304	0.224	0.310	0.217	0.320	0.228	0.317	0.214	0.302
	720	**0.230**	**0.321**	0.231	0.326	0.238	**0.321**	0.251	0.340	0.265	0.341	0.284	0.363	0.270	0.348	0.254	0.333
ETTh1	96	0.388	0.405	0.389	0.408	0.390	0.408	0.390	0.411	**0.382**	**0.399**	0.478	0.448	0.389	0.400	0.396	0.413
	192	0.438	0.436	0.432	0.432	**0.428**	0.432	0.432	0.439	0.431	**0.426**	0.561	0.504	0.440	0.43	0.458	0.448
	336	0.478	0.455	0.475	0.451	**0.462**	0.451	0.480	0.460	0.476	**0.449**	0.612	0.537	0.482	0.453	0.508	0.472
	720	**0.483**	**0.472**	0.515	0.492	0.489	0.476	0.532	0.500	0.495	0.487	0.601	0.541	0.486	0.479	0.546	0.503
Exchange	192	0.231	0.340	0.210	0.330	0.239	0.342	0.221	0.337	**0.175**	**0.297**	0.259	0.370	0.178	0.301	0.177	0.300
	336	0.431	0.472	0.387	0.451	0.479	0.486	0.387	0.453	**0.322**	**0.409**	0.478	0.501	0.328	0.415	0.326	0.414
ILI	60	2.02	**0.885**	2.15	0.923	2.48	0.944	2.45	0.994	**1.99**	0.905	2.37	0.966	2.31	0.970	1.90	0.868
Traffic	96	0.486	**0.322**	**0.483**	0.324	0.496	0.325	0.502	0.330	0.606	0.389	0.611	0.336	0.643	0.405	0.524	0.351
	192	**0.492**	**0.325**	0.500	0.330	0.497	0.327	0.523	0.331	0.592	0.382	0.643	0.352	0.603	0.387	0.519	0.346
	336	**0.506**	0.331	0.520	0.337	0.509	**0.328**	0.552	0.338	0.600	0.384	0.662	0.363	0.612	0.389	0.530	0.350
	720	**0.523**	**0.340**	0.575	0.362	0.525	0.350	0.626	0.369	0.633	0.401	0.678	0.365	0.652	0.406	0.562	0.366
Weather	96	0.165	**0.211**	0.166	0.219	**0.161**	**0.211**	0.175	0.214	0.193	0.232	0.169	0.220	0.194	0.233	0.182	0.222
	192	**0.210**	**0.254**	0.216	0.261	0.212	0.255	0.226	0.266	0.238	0.269	0.223	0.264	0.238	0.268	0.228	0.261
	336	**0.266**	**0.294**	0.273	0.300	**0.266**	0.295	0.280	0.303	0.291	0.306	0.279	0.302	0.290	0.304	0.282	0.299
	720	**0.342**	**0.343**	0.350	0.349	0.343	0.344	0.352	0.350	0.365	0.354	0.359	0.355	0.363	0.35	0.359	0.349
Best Count (/20)		8	11	4	2	5	4	0	0	4	5	0	0	0	0	-	-
Average		**0.445**	**0.382**	0.452	0.384	0.469	0.386	0.478	0.393	0.466	0.394	0.525	0.412	0.488	0.401	0.449	0.386

Table D.2: Results of multi-task forecasting.

18 Tasks	Shared (1 model)				Task-specific (18 models)					
	UniTS + CM		UniTS		iTransformer	TimesNet	PatchTST	Pyraformer	Autoformer	GPT4TS
	Sup.	PT	Sup.	PT	Sup.					FT
Heartbeat	67.3	70.2	59.0	69.3	66.8	**72.7**	65.9	**72.7**	71.7	69.8
JapaneseVowels	94.1	93.2	93.5	90.8	95.9	**97.6**	94.1	85.4	94.1	94.6
PEMS-SF	83.2	82.1	83.2	85.0	83.2	77.5	**83.8**	83.2	79.2	79.2
SelfRegulationSCP2	**58.3**	51.7	47.8	53.3	48.9	52.8	48.9	56.7	45.0	45.6
SpokenArabicDigits	97.1	93.5	97.5	92.0	97.8	**98.7**	97.5	92.1	97.3	97.5
UWaveGestureLibrary	**84.4**	83.8	79.1	75.6	82.2	**84.4**	81.9	72.2	42.2	81.9
ECG5000	93.4	93.4	92.6	93.4	93.3	92.6	**94.3**	91.4	91.9	93.0
NonInvasiveFetalECGThorax1	89.5	55.2	**90.5**	27.1	88.2	88.9	86.5	21.4	21.7	89.7
Blink	**99.1**	95.6	**99.1**	91.1	93.3	87.6	89.6	88.2	63.1	92.4
FaceDetection	64.7	54.6	64.1	57.6	66.0	66.2	63.9	**67.3**	59.2	66.1
ElectricDevices	62.4	60.5	60.3	55.4	57.3	49.5	59.5	**65.4**	56.1	62.9
Trace	**99.0**	93.0	91.0	82.0	79.0	91.0	77.0	74.0	60.0	96.0
FordB	**76.2**	64.2	76.0	62.8	72.7	68.9	61.4	55.3	66.4	77.7
MotionSenseHAR	92.8	**94.3**	92.8	93.2	93.6	90.6	75.8	88.7	30.2	96.2
EMOPain	75.5	80.8	78.0	80.3	79.4	78.0	79.2	**81.4**	69.9	79.4
Chinatown	97.7	**98.0**	97.7	**98.0**	97.4	97.7	97.7	27.4	96.8	96.5
MelbournePedestrian	89.3	78.3	87.3	77.0	89.3	**95.7**	80.4	52.3	75.0	94.0
SharePriceIncrease	62.9	66.6	61.9	**68.4**	61.9	65.0	68.0	63.1	61.5	63.7
1st Count (/18)	5	2	2	2	0	5	2	4	0	-
2nd Count (/18)	6	5	3	1	5	2	2	2	1	-
Average Score	**82.0**	78.3	80.6	75.1	80.3	80.9	78.1	68.8	65.6	82.0

Table D.3: Results of multi-task classification.

D.2 Few-shot Learning

For the few-shot tasks, we conduct four distinct tasks: forecasting (FCST), classification (CLS), imputation (IMP), and anomaly detection (AD), which are discussed in Sections D.2.1, D.2.2, D.2.3, and D.2.4, respectively.

D.2.1 Few-shot Forecasting

The results of few-shot forecasting with data ratios of 5%, 15%, and 20% are shown in Tables D.4, D.5, and D.6, respectively.

5%		iTransformer		UniTS				UniTS + CM			
		FT		PT		FT		PT		FT	
Data	H	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh2	96	0.554	0.500	**0.405**	**0.417**	0.418	0.424	0.421	0.427	0.421	0.425
	192	0.440	0.438	0.400	0.406	0.377	0.397	0.386	0.402	**0.370**	**0.389**
	336	0.478	0.467	0.425	0.433	0.420	0.433	0.423	0.431	**0.416**	**0.425**
	720	0.483	0.480	0.446	0.457	0.439	0.452	**0.424**	0.444	0.428	**0.443**
RiverFlow	24	1.141	0.514	1.115	0.504	1.112	0.504	**1.097**	0.503	**1.097**	**0.500**
ETTm1	96	0.504	0.462	0.436	0.434	0.384	0.404	0.428	0.436	**0.354**	**0.384**
	192	0.555	0.485	0.462	0.448	0.414	0.418	0.475	0.458	**0.393**	**0.405**
	336	0.567	0.496	0.560	0.494	0.453	0.442	0.550	0.493	**0.420**	**0.423**
	720	0.659	0.539	0.703	0.558	0.526	0.483	0.689	0.554	**0.483**	**0.455**
Average		0.598	0.487	0.549	0.461	0.505	0.440	0.546	0.462	**0.489**	**0.429**

Table D.4: Results of few-shot forecasting (5%).

15%		iTransformer		UniTS				UniTS + CM			
		FT		PT		FT		PT		FT	
Data	H	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh2	96	0.441	0.440	0.403	0.412	**0.399**	**0.409**	0.416	0.423	0.403	0.411
	192	0.398	0.410	0.396	0.404	0.394	0.399	0.388	0.403	**0.387**	**0.399**
	336	0.436	0.441	0.432	0.435	0.441	0.435	**0.419**	0.435	0.430	**0.431**
	720	0.438	0.453	0.448	0.457	0.449	0.453	**0.415**	**0.442**	0.433	0.446
RiverFlow	24	**1.067**	**0.467**	1.077	0.492	1.069	0.489	1.073	0.492	1.072	0.487
ETTm1	96	0.423	0.419	0.407	0.420	0.353	0.386	0.408	0.426	**0.342**	**0.380**
	192	0.464	0.439	0.434	0.432	0.384	0.400	0.449	0.447	**0.377**	**0.399**
	336	0.492	0.457	0.490	0.464	0.416	0.420	0.502	0.475	**0.406**	**0.148**
	720	0.558	0.493	0.641	0.537	0.480	0.455	0.621	0.530	**0.470**	**0.451**
Average		0.524	0.450	0.525	0.450	0.487	0.428	0.522	0.452	**0.481**	**0.425**

Table D.5: Results of few-shot forecasting (15%).

20%		iTransformer		UniTS				UniTS + CM			
		FT		PT		FT		PT		FT	
Data	H	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh2	96	0.418	0.426	0.411	0.414	**0.391**	**0.405**	0.411	0.422	0.395	0.409
	192	0.395	0.407	0.383	**0.398**	0.395	0.403	**0.381**	0.400	0.390	0.400
	336	0.431	0.438	**0.419**	0.431	0.430	**0.430**	0.423	**0.430**	0.438	0.433
	720	0.431	0.449	0.440	0.453	0.444	0.449	**0.418**	**0.422**	0.456	0.456
RiverFlow	24	**1.056**	**0.462**	1.069	0.487	1.069	0.489	1.071	0.487	1.067	0.489
ETTm1	96	0.408	0.410	0.409	0.421	0.344	0.379	0.403	0.425	**0.339**	**0.376**
	192	0.444	0.428	0.443	0.439	0.377	0.397	0.450	0.450	**0.375**	**0.396**
	336	0.471	0.445	0.505	0.472	0.408	0.418	0.507	0.481	**0.403**	**0.415**
	720	0.536	0.482	0.648	0.536	0.472	0.453	0.621	0.531	**0.466**	**0.448**
Average		0.510	0.438	0.525	0.450	0.486	0.425	0.521	0.453	**0.482**	**0.425**

Table D.6: Results of few-shot forecasting (20%).

D.2.2 Few-shot Classification

The results of few-shot classification with data ratios of 5%, 15%, and 20% are shown in Tables D.7, D.8, and D.9, respectively.

5%	iTransformer	UniTS		UniTS + CM	
	FT	PT	FT	PT	FT
ECG200	78.0	67.0	77.0	**80.0**	77.0
Handwriting	5.4	4.6	4.7	4.8	**5.5**
SelfRegulationSCP1	62.8	66.2	74.7	**77.8**	73.7
RacketSports	37.5	31.6	35.5	39.5	**47.4**
Epilepsy	39.9	44.9	47.1	44.9	**57.2**
StarLightCurves	85.1	82.3	83.8	**86.3**	85.4
Average	51.4	49.4	53.8	**54.9**	54.8

Table D.7: Results of few-shot classification (5%).

15%	iTransformer	UniTS		UniTS + CM	
	FT	PT	FT	PT	FT
ECG200	81.0	74.0	78.0	73.2	**82.0**
Handwriting	**9.8**	7.3	8.1	9.2	8.5
SelfRegulationSCP1	67.9	59.0	**76.5**	69.3	68.6
RacketSports	**54.6**	40.1	50.7	44.7	51.3
Epilepsy	41.3	52.9	58.0	61.6	**68.1**
StarLightCurves	84.2	85.8	**87.1**	85.9	85.5
Average	56.5	53.2	59.7	55.4	**60.4**

Table D.8: Results of few-shot classification (15%).

20%	iTransformer	UniTS		UniTS + CM	
	FT	PT	FT	PT	FT
ECG200	81.0	76.0	77.0	**85.0**	82.0
Handwriting	**11.8**	8.0	8.5	7.6	9.8
SelfRegulationSCP1	77.1	68.6	70.6	**77.8**	74.4
RacketSports	54.6	51.3	**57.9**	38.8	50.7
Epilepsy	62.3	81.9	72.5	**84.1**	61.6
StarLightCurves	84.8	87.3	86.0	**90.0**	87.8
Average	59.9	58.9	63.6	60.0	**64.8**

Table D.9: Results of few-shot classification (20%).

D.2.3 Few-shot Imputation

The results of few-shot imputation with data ratios of 25% and 50% are shown in Table D.10

Ratio			ECL	ETTh1	ETTh2	ETTm1	ETTm2	Weather	Avg.
25%	TimesNet	FT	0.245	0.369	0.193	0.442	0.119	0.106	0.246
	PatchTST	FT	0.195	0.315	0.147	0.309	0.092	0.089	0.191
	iTransformer	FT	0.174	0.301	0.185	0.254	0.113	0.087	0.186
	UniTS	PT	0.139	0.311	0.178	0.268	0.102	0.078	0.179
		FT	0.160	0.284	0.150	0.241	**0.090**	0.077	0.167
	UniTS + CM	PT	0.139	0.310	0.176	0.262	0.100	0.078	0.179
		FT	**0.129**	**0.275**	**0.149**	**0.231**	**0.090**	**0.073**	**0.158**
50%	TimesNet	FT	0.258	0.412	0.211	0.607	0.140	0.125	0.292
	PatchTST	FT	0.230	0.353	0.175	0.442	**0.111**	0.105	0.236
	iTransformer	FT	0.203	0.332	0.205	0.372	0.136	0.106	0.226
	UniTS	PT	0.172	0.352	0.251	0.380	0.134	0.103	0.232
		FT	0.191	0.322	0.198	0.352	0.118	0.095	0.213
	UniTS + CM	PT	0.162	0.353	0.240	0.370	0.128	0.097	0.225
		FT	**0.151**	**0.307**	**0.197**	**0.345**	0.116	**0.093**	**0.201**

Table D.10: Results of few-shot imputation.

D.2.4 Few-shot Anomaly Detection

The results of few-shot anomaly detection with data ratio of 5% are shown in Table D.11.

		MSL	PSM	SMAP	SMD	SWAT	Avg.
Anomaly Trans.	-	78.0	90.2	68.3	77.8	81.5	79.2
TimesNet	FT	33.9	91.0	68.5	84.0	**93.4**	74.2
iTransfomer	FT	80.4	96.5	67.2	82.4	89.0	83.1
PatchTST	FT	79.9	96.6	68.7	83.8	92.6	84.3
UniTS	PT	73.2	95.5	65.9	81.2	92.9	81.7
	FT	**81.3**	**97.3**	71.6	85.5	92.5	85.6
UniTS + CM	PT	73.7	95.5	66.0	82.0	92.9	82.0
	FT	**81.3**	**97.3**	**75.9**	**86.2**	92.6	**86.6**

Table D.11: Results of few-shot anomaly detection.

E Application to TimeSiam

To demonstrate the effectiveness of our proposed model on TimeSiam [21], which uses a self-supervised pretraining framework for TS with Siamese networks, we conduct experiments using iTransformer [10] as the backbone, with two datasets that vary in channel size: Exchange, with a small number of channels (8), and ECL, with a large number of channels (321). Specifically, we apply variants of our method by using the domain parameter only during the fine-tuning stage and during both pretraining and fine-tuning stages. The results, shown in Table E.1, validate both components of our method, with the best performance achieved when using domain parameters at both pretraining and fine-tuning stages.

		TimeSiam		+ CM					
Correlation matrix		-		✓		✓		✓	
Domain parameters	Pretrain	-		-		-		✓	
	Fine-tune	-		-		✓		✓	
Dataset	H	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Exchange ($C=8$)	96	0.092	0.215	<u>0.089</u>	**0.207**	**0.088**	**0.207**	**0.088**	<u>0.209</u>
	192	**0.182**	0.306	**0.182**	<u>0.304</u>	**0.182**	**0.303**	**0.182**	0.305
	336	0.341	0.426	0.336	<u>0.422</u>	<u>0.332</u>	**0.417**	**0.329**	**0.417**
	720	0.806	0.679	0.792	0.670	<u>0.788</u>	<u>0.668</u>	**0.783**	**0.666**
	Avg.	0.356	0.407	0.350	0.401	<u>0.349</u>	<u>0.399</u>	**0.346**	**0.398**
ECL ($C=321$)	96	0.147	0.239	**0.140**	**0.236**	**0.140**	**0.236**	<u>0.141</u>	<u>0.237</u>
	192	0.162	0.253	**0.157**	<u>0.251</u>	**0.157**	<u>0.251</u>	**0.157**	**0.250**
	336	0.175	0.269	<u>0.173</u>	<u>0.268</u>	<u>0.173</u>	<u>0.268</u>	**0.172**	**0.267**
	720	0.215	0.304	**0.203**	<u>0.297</u>	**0.203**	<u>0.297</u>	**0.203**	**0.296**
	Avg.	0.175	0.266	**0.168**	<u>0.263</u>	**0.168**	<u>0.263</u>	**0.168**	**0.262**

Table E.1: Results of TS forecasting with TimeSiam.

F Application to Moirai

Although Moirai [13] is a CI method, CM can still be adapted to it, as Moirai performs channel flattening followed by time-axis attention. The CM can be applied by reshaping it from $C \times C$ to $C \cdot P \times C \cdot P$ with duplicated intra-channel values, where P is the number of patches. However, we exclude it for the following reasons:

- **Limited task coverage**: Moirai supports only forecasting task, whereas UniTS handles four tasks.
- **Reproducibility issues**: Its code lacks pretraining details, and our reproduced fine-tuning results were significantly worse than reported, which is also shared with many users in the official GitHub.
- **Resource constraints**: Moirai's large data (231B) and model size (91M) make experiments infeasible under our resource limits.

G Lookback Window Size vs. Performance

Following the previous work [10], we conduct an experiment to evaluate the effect of varying the lookback window size (L) on performance, using three datasets: ECL [29], Traffic [29], and PEMS03 [31] with iTransformer [10] as the backbone. The results, shown in Figure G.1, indicate that the effectivness of CM remains robust to the choice of L for all three datasets.

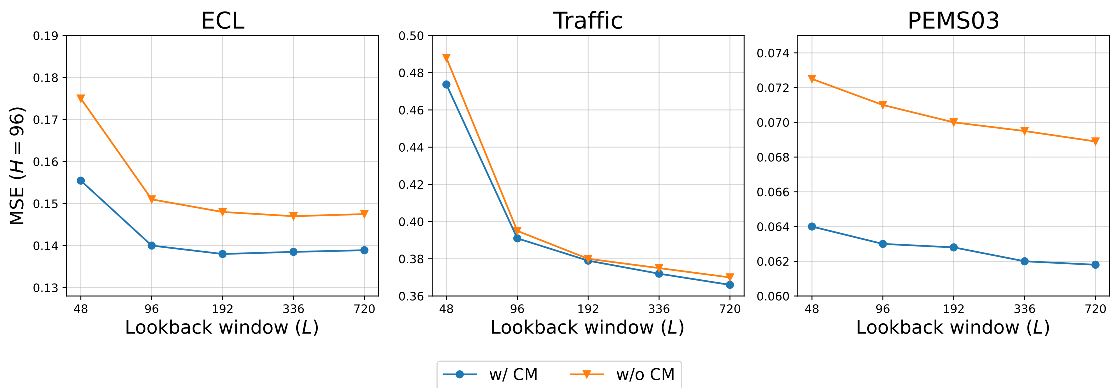

Fig. G.1: **Effect of CM under various lookback window sizes.** Forecasting performance with the lookback length $L \in \{48, 96, 192, 336, 720\}$, with forecast horizon $H = 12$ for PEMS03 and $H = 96$ for other datasets.

H CM under Extreme Cases

To evaluate the effectiveness of CM under extreme cases, we design a scenario where the channels in TS exhibit no correlation. Specifically, we generate a synthetic TS dataset with two channels using sine waves oscillating at frequencies of 0.5 and 2.0 over a length of 18,000 (similar to ETTh [20]), as shown in Figure H.1. We conduct TS forecasting using this dataset with iTransformer [10] as the backbone, with an input window size and forecasting horizon of 96, following the experimental protocol used in ETTh1. The result yields a CD ratio of CM approximately 0.018 and a forecasting MSE of around 0.0014, confirming strong channel independence and demonstrating the effectiveness of our method even under extreme CI conditions.

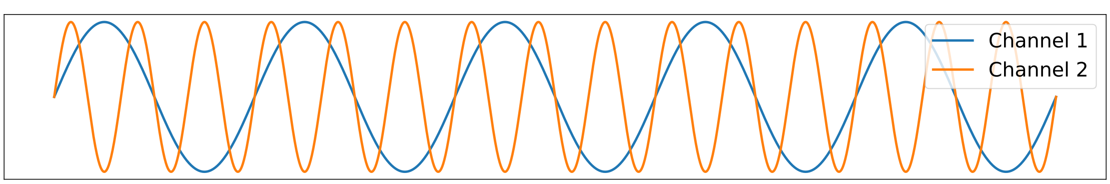

Fig. H.1: Synthetic dataset with two uncorrelated channels

I Masked Channel Prediction

Tables I.1 and I.2 show the results of masked channel prediction for five datasets [29, 31], indicating significant improvement when a CM is applied to iTransformer compared to when it is not used.

Horizon	Exchange			ECL		
	Avg. MSE(C1∼C8)			Avg. MSE(C1∼C321)		
	iTrans.	+ CM	Imp.	iTrans.	+ CM	Imp.
96	0.139	**0.138**	**1.2%**	0.846	**0.526**	**37.8%**
192	0.236	**0.232**	**1.5%**	0.849	**0.563**	**33.7%**
336	0.383	**0.374**	**2.4%**	0.861	**0.594**	**31.0%**
720	0.934	**0.917**	**1.8%**	0.891	**0.741**	**16.8%**
Avg.	0.423	**0.415**	**1.8%**	0.862	**0.606**	**29.7%**

Table I.1: Results of masked channel prediction (Exchange, ECL).

Horizon	PEMS04			PEMS07			PEMS08		
	Avg. MSE(C1∼C307)			Avg. MSE(C1∼C883)			Avg. MSE(C1∼C170)		
	iTrans.	+ CM	Imp.	iTrans.	+ CM	Imp.	iTrans.	+ CM	Imp.
12	0.549	**0.300**	**45.4%**	0.835	**0.343**	**58.9%**	0.628	**0.200**	**68.1%**
24	0.718	**0.351**	**51.1%**	0.865	**0.448**	**48.1%**	0.678	**0.241**	**64.5%**
48	0.750	**0.409**	**45.5%**	1.038	**0.511**	**50.8%**	1.197	**1.059**	**11.5%**
96	0.758	**0.513**	**32.3%**	1.040	**0.640**	**38.5%**	1.375	**1.217**	**11.5%**
Avg.	0.694	**0.393**	**43.3%**	0.945	**0.486**	**48.6%**	0.970	**0.679**	**29.9%**

Table I.2: Results of masked channel prediction (PEMS datasets).